Artificial Intelligence, Data and Creative Destruction*

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Abstract

Artificial intelligence (AI) can codify tacit knowledge, but what it codifies inside a firm is that firm's own knowledge. I build and calibrate an endogenous growth model in which firms implement the same technology to different degrees, governed by their accumulated data. Implementation substitutes capital for labour and relaxes the span of control, since tasks performed by AI need no oversight, so how efficiency scales with firm size becomes an equilibrium outcome. Because the return to implementation rises with data, which are lost on exit, AI favours large incumbents, makes them harder to displace and reduces the number of firms. Growth rises as incumbents expand, but weaker creative destruction bounds the gain: doubling AI's potential raises growth by only 0.26 percentage points. Requiring firms to release their data after a few years turns them into public, accumulating knowledge and outperforms research and entry subsidies on welfare, at no fiscal cost.

Keywords: artificial intelligence, firm data, creative destruction, span of control, firm dynamics, data disclosure

JEL Classification: O33, O41, O31, L11, L25, O38.

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1. Introduction

A machine can perform a task only once the knowledge of how the task is done has been *codified* into a form the machine can run. Earlier automation codified by articulation, since someone had to state the rules in advance, and this confined it to routine tasks (Autor *et al.*, 2003; Autor, 2014; Acemoglu and Restrepo, 2020). Artificial intelligence (AI) learns the knowledge embedded in how a task has been done without anyone articulating it, so it can write code, synthesise information and solve unstructured problems. These are also the tasks that have historically bounded how large a firm can grow, since scarce managerial time, talent and the cost of transmitting knowledge inside the organisation limit the span of control (Penrose, 1959; Lucas, 1978; Garicano, 2000).

Firms can train AI on two sources of knowledge, public knowledge and their own. The first is common to every firm and confers no advantage. The second is where firms differ, in the tacit knowledge they hold about their own processes (Atkeson and Kehoe, 2005) and in the data that record them (Jones and Tonetti, 2020; Farboodi and Veldkamp, 2026; Baley and Veldkamp, 2025), and neither moves easily between firms. TSMC holds years of sensor and yield data from its own fabrication lines, together with the process adjustments its engineers have learnt to make, and its rivals can observe neither. Amazon holds a record of how goods move through its fulfilment network that no other retailer can reconstruct. AI allows these firms to codify what they know and apply it at scale. In the cleanest evidence so far, an assistant trained on one firm's own customer conversations passed the practices of its most productive agents to the rest of its workforce, and the least experienced agents gained the most (Brynjolfsson *et al.*, 2025). A rival can license the same model, but not the conversations it learned from.

The technology therefore lowers the cost of moving knowledge within the firm while leaving the cost of moving it between firms largely intact, so the margin that matters is not whether a firm has adopted it but how far it has got. Its use is concentrated in large firms (McElheran *et al.*, 2024; Eurostat, 2025). In the United States the 18 per cent of firms that use it employ 32 per cent of workers (Bonney *et al.*, 2026), and firms that invest more in it

grow faster, the more so the larger they are, while the industries that invest most become more concentrated (Babina *et al.*, 2024). I build an endogenous growth model in which firms implement AI to different degrees, governed by the data they have accumulated, and I use it to trace the consequences for competition, firm dynamics and growth, and to ask how policy should respond.

Part of what AI can do needs only an off-the-shelf model, and every firm implements this public part. How far a firm implements the rest depends on its data, the records of how its processes run, which grow with the number of products it runs (Farboodi *et al.*, 2019). Implementation works through two channels. The first substitutes a cheap factor for an expensive one, the channel the literature has studied (Zeira, 1998; Acemoglu and Restrepo, 2018; Aghion *et al.*, 2019; Acemoglu and Restrepo, 2022). The second changes how the firm is managed (Ide and Talamàs, 2025; Garicano *et al.*, 2026), and since the firm's capacity to control its processes bounds the span of control, it changes how many products the firm can run.

Implementation and data both lower a firm's marginal cost of production, and marginal cost decides who wins the lines that change hands. A firm takes a line by innovating on it, and it wins only if its quality step outweighs the incumbent's cost advantage. Firms with more data therefore win more of the lines their innovations reach and keep their own for longer, and each line they win adds to their data. An entrant enters in the same way, by taking a line with an innovation, but it starts with the least data in the economy, so the incumbent's cost advantage over it is the barrier to entry. Since the firms with the most data implement the largest share of any advance, every advance widens that advantage and raises the barrier.

A higher barrier is only half of the story, and AI sets two forces against each other. It raises incumbents' profits and cuts their cost of research by more than that of entrants, so incumbents innovate more and displace one another more often, which raises growth. At the same time entrants win fewer of the contests they enter, so creative destruction shifts from new firms to incumbents and fewer lines change hands for a given amount of research.

The source of the incumbents' advantage points to a remedy. Unlike the knowledge in an innovation, which every later producer builds on, the knowledge in records stays with the firm that collected them and is lost when it exits. I study a rule that makes every firm's records public after a fixed number of years. Public records serve every firm and entrant as data, and their content accumulates in the knowledge every producer builds on, so disclosure turns data into transferable and accumulable knowledge. At the potential of the optimistic forecast, disclosure after ten years raises growth from 1.60 to 2.03 per cent and welfare by 19 per cent at no fiscal cost, whereas the taxes and subsidies that achieve the same growth raise welfare by 3 to 11 per cent. Because records date quickly, the lag must be far shorter than a patent term.

Related literature. A large literature models automation as the substitution of capital for labour in a growing set of tasks (Zeira, 1998; Acemoglu and Restrepo, 2018; Acemoglu and Restrepo, 2022) and measures its effects and the exposure of occupations to AI (Autor *et al.*, 2003; Acemoglu and Restrepo, 2020; Webb, 2020; Eloundou *et al.*, 2023). Exposure describes what a technology could do, not what a given firm has done with it. I keep the displacement mechanism and show that the gap between exposure and implementation is systematic, because it depends on data, which also govern firm size and survival.

A more recent literature studies how AI reshapes the organisation of work. In Ide and Talamàs (2025) it enables management by exception, expanding the span of control of the most knowledgeable and contracting that of the least, and Garicano *et al.* (2026) argue that it displaces jobs only where the bundle of tasks held together by judgement, coordination and tacit knowledge is weak. In my model the span of control runs over a firm's product lines rather than over workers in one hierarchy, and the data that AI learns from stay inside the firm, which makes AI a force for incumbent growth, concentration and entry deterrence.

Managerial capacity bounds the size of the firm (Penrose, 1959; Lucas, 1978), including in hierarchies of authority and knowledge (Rosen, 1982; Garicano, 2000; Garicano and Rossi-Hansberg, 2006; Caliendo and Rossi-Hansberg, 2012), shared know-how bounds its

scope (Teece, 1980), and information technology relaxes these bounds (Bloom *et al.*, 2014). In this literature technology changes the span of control over workers. In mine it changes the span of control over a firm's product lines, and the elasticity of efficiency with respect to the number of lines rises with the technology (Proposition 1), which brings the span of control into the firm dynamics of Klette and Kortum (2004).

The closest paper is Beraja and Talamàs (2026), in which AI also codifies tacit organisational knowledge but makes the knowledge of mature firms portable, so entrants are born old and the gain is a level effect. I make the opposite assumption. Beyond its public part, AI can be implemented only with data that stay with the firm, and implementation changes the production technology, so the firms with the most data gain most, concentration rises and the mass of firms falls. Disclosure moves the economy towards theirs.

A growing literature links intangible capital and information technology to rising market power and declining business dynamism (Akcigit and Ates, 2021; Crouzet *et al.*, 2022; Aghion *et al.*, 2023; Lashkari *et al.*, 2024; Chiavari and Goraya, 2025). In Aghion *et al.* (2023) information technology lowers growth by letting the most efficient firms expand, as AI does in my model once it is far advanced. In De Ridder (2024) software raises fixed costs and lowers marginal costs, which deters entry, and the intangible is rival across a firm's products. In my model the intangible, the firm's data, is non-rival across its lines, so a firm's automation intensity in one line rises with its size, which gives a direct test between the two.

Work on AI and aggregate growth asks whether AI can sustain explosive growth (Aghion *et al.*, 2019), shows that Uzawa's theorem (Uzawa, 1961; Grossman *et al.*, 2017) can hold even when all technical progress is embodied in capital (Jones and Liu, 2024), and finds that the tasks that resist automation become the weak link on aggregate output (Jones and Tonetti, 2026). Acemoglu (2025) estimates modest gains because AI can profitably take over only a limited share of tasks, and Aghion and Bunel (2024) bound the gains between his values and far larger ones. I add a limit that works through firm dynamics, since the entry that AI discourages removes part of the growth that incumbents' expansion would otherwise deliver.

Finally, data are non-rival, so their broad use has large social gains, but firms that fear creative destruction hoard them (Jones and Tonetti, 2020), large firms accumulate more of them (Farboodi *et al.*, 2019), and data-driven markets tip towards monopoly unless competitors share their data (Prüfer and Schottmüller, 2021), as the Digital Markets Act now requires of the largest search engines (European Parliament and Council of the European Union, 2022). Patents trade exclusive rights for disclosure, which raises follow-on innovation (Furman *et al.*, 2021; Hegde *et al.*, 2023). I study a rule of the same kind for business records, with a lag set by the life of records rather than of inventions.

2. Theoretical Model

2.1. Economic Environment

Time is continuous and indexed by t. A representative household has logarithmic preferences over consumption,

$$\int_0^\infty e^{-\rho t} \ln C_t \, dt, \tag{1}$$

where C_t is consumption and $\rho > 0$ is the rate of time preference. The household supplies one unit of labour inelastically, owns the equity of every firm in the economy, and faces the flow budget constraint

$$\dot{A}_t = r_t A_t + w_t - C_t, \tag{2}$$

where A_t denotes assets, w_t the wage and r_t the interest rate, subject to the usual no-Ponzi condition. The consumption good is the numeraire. Since the household holds the entire equity of the economy, assets equal the aggregate value of firms,

$$A_t = \int_{\mathcal{F}_t} V_{it} \, di, \tag{3}$$

where $\mathcal{F}_t$ is the set of active firms and V_{it} is the value of firm i.

The final good is produced competitively from a continuum of production lines $j \in$

$[0,1]$ with a constant elasticity of substitution,

$$Y_t = \left(\int_0^1 y_{jt}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}}, \qquad \varepsilon > 1, \tag{4}$$

and its price is normalised to one. Each line is produced by one firm, its incumbent, which is the only producer of that line. Firm i may operate more than one line. Its set of active lines is $\mathcal{J}_{it} \subseteq [0,1]$, and $n_{it} = |\mathcal{J}_{it}|$ is the number of lines it operates.

Output of firm i on line j combines a continuum of task types $x \in [0,1]$,

$$y_{ijt} = \Psi(d_{it}, n_{it}, I_{it}) \left(\int_0^1 y_{ijt}(x)^{\frac{\sigma-1}{\sigma}} \, dx \right)^{\frac{\sigma}{\sigma-1}}, \tag{5}$$

where $\Psi(d_{it}, n_{it}, I_{it})$ is control efficiency, which depends on the firm's data intensity d, its number of lines n and implementation I. The elasticity of substitution across tasks is $\sigma < 1$, so tasks are gross complements. Tasks are types rather than instances, so a firm operating n lines performs every task type in each of them. Each task is performed either by capital or by labour,

$$y_{ijt}(x) = \begin{cases} \gamma_k \, k_{ijt}(x), & x \le I_{it}, \\ \gamma_\ell \, \bar{Q}_t \, \ell_{ijt}(x), & x > I_{it}, \end{cases} \tag{6}$$

where $k_{ijt}(x)$ is capital input, $\ell_{ijt}(x)$ is labour input, and I_{it} is the firm's implementation, defined below: task types up to I_{it} are performed by capital and the rest by labour. The productivity of labour is proportional to $\bar{Q}_t > 0$, the *leading edge*, which is the stock of codified knowledge in the economy at date t. Codified knowledge has been made explicit, so it is not tied to the firm that created it, and the leading edge is a single economy-wide index, common to every producer and every line. It starts from a given $\bar{Q}_0 > 0$ and grows only through innovation, since every innovation that captures a line adds its step to it, as set out in (26) below. A worker's productivity in any task type is therefore $\gamma_\ell \bar{Q}_t$, with $\gamma_\ell > 0$, the same in every firm, and it rises as knowledge accumulates. Capital consists of machines, produced at a constant marginal cost of ψ units of the final good, and a machine performs any task type it can perform with productivity γ_k. With the final good

as numeraire, a machine task costs ψ/γ_k and a labour task costs $w_t/(\gamma_\ell \bar{Q}_t)$, so the relative price of a machine task is

$$\kappa_t \equiv \frac{\psi/\gamma_k}{w_t/(\gamma_\ell \bar{Q}_t)}. \tag{7}$$

Knowledge makes workers more productive and leaves machines as they are, so technical change is labour-augmenting.

Assumption 1 (Artificial intelligence is cost saving)**.** Performing a task type with capital costs less than performing it with labour: $\kappa_t < 1$ for all t.

Assumption 1 makes implementation worthwhile wherever it is possible. A firm therefore assigns every implemented task type to capital and every other task type to labour, in production and, as set out below, in research. Let $\bar{I}_t \in [0, 1)$ denote the *potential*, the measure of task types that artificial intelligence can perform. It is exogenous, economy-wide and common to every firm, and it is the sense in which artificial intelligence is a shock to the economy rather than to any particular firm. *Implementation* $I_{it} \leq \bar{I}_t$ is the measure of task types on which firm i has actually deployed it. The potential has two parts. A share $\zeta \in [0, 1)$ of it, the *public part*, consists of task types that an off-the-shelf model performs with what it has learned from public data, so every producer implements it. The remaining share $1 - \zeta$, the *firm-specific part*, consists of task types that a model performs only once it has been adapted to how the firm itself works, and how much of it a firm implements is governed by the firm's own data, as set out below. The potential is the interval $[0, \bar{I}_t]$, its public part is $[0, \zeta \bar{I}_t]$, and firm i has implemented $[0, I_{it}]$.

A firm's *data intensity* d_{it} is how much of the operations on each of its lines it measures and records, in transaction and inventory records, quality and maintenance logs, records of customer interactions and performance indicators. Recording is a firm-wide practice, so it applies to every line the firm runs, and data are a by-product of production (Farboodi *et al.*, 2019), so the firm's data grow with the number of its lines. I write the firm's data as

$$D_{it} \equiv \left(d_{it}\, n_{it}\right)^{\eta}, \qquad \eta > 0, \tag{8}$$

where η governs the returns to data. Data are non-rival within the firm, but firms do not share them (Jones and Tonetti, 2020), and the records of how one firm's processes run are of little use to another. Data play two distinct roles. First, they govern how much of the firm-specific part of the potential the firm can put into practice. Artificial intelligence is general-purpose, but outside the public part a system has to be adapted to how a task is actually done inside the firm that deploys it, as when a customer-service assistant is fine-tuned on the firm's own records of customer conversations (Brynjolfsson *et al.*, 2025), and that adaptation draws on exactly these data. Second, they raise the efficiency with which the firm controls its processes.

Assumption 2 (Implementation capacity)**.** Firm i implements the whole public part and a share $1 - e^{-\varsigma D_{it}}$ of the firm-specific part,

$$I_{it} \;=\; \zeta\,\bar{I}_t + (1-\zeta)\big(1 - e^{-\varsigma D_{it}}\big)\,\bar{I}_t \;=\; \mathcal{A}(D_{it})\,\bar{I}_t, \tag{9}$$

where $\mathcal{A}(D) \equiv 1 - (1-\zeta)\,e^{-\varsigma D} \in (\zeta, 1)$, with $\varsigma > 0$, is the firm's *implementation capacity*.

Implementation moves when the firm's data grow, through its data intensity or its number of lines, and when the potential advances, and either way the task types added at the top of $[0, I_{it}]$ pass from labour to capital.

Lemma 1 (Properties of implementation)**.** *Under* (8) *and* (9)*, for every firm and date: (i)* $I_{it} < \bar{I}_t$ *whenever* $\bar{I}_t > 0$*, with shortfall* $\bar{I}_t - I_{it} = (1-\zeta)\,e^{-\varsigma D_{it}}\,\bar{I}_t$*; (ii)* $\partial I_{it}/\partial \bar{I}_t = \mathcal{A}(D_{it}) \in (\zeta, 1)$*, increasing in* d_{it} *and in* n_{it}*; (iii)* $\partial^2 I_{it}/\partial d_{it}\,\partial \bar{I}_t \;=\; \mathcal{A}'(D_{it})\,\eta D_{it}/d_{it} \;>\; 0$ *and* $\partial^2 I_{it}/\partial n_{it}\,\partial \bar{I}_t \;=\; \mathcal{A}'(D_{it})\,\eta D_{it}/n_{it} > 0$.

Every firm implements the public part in full, so firms differ only in how much of the firm-specific part they implement, and the shortfall in (i) lies entirely in that part. By (ii), two firms with data $D > D'$ implement shares of an advance in the potential that differ by $\mathcal{A}(D) - \mathcal{A}(D') = (1-\zeta)\big(e^{-\varsigma D'} - e^{-\varsigma D}\big)$, so the larger the public share, the less an advance separates firms in what they implement. Data grow with the lines a firm runs, so larger firms implement more, which is the size gradient in the use of artificial intelligence

across firms (Bonney *et al.*, 2026). Appendix A proves the lemmas and propositions of this section.

A firm operating n lines with implementation I has to control the tasks that artificial intelligence does not perform, in each of its lines.[1] Its *span of control* is

$$S(n, I) = (1 - I)\, n, \tag{10}$$

the measure of processes that people still run and that the firm has to monitor, coordinate and correct. It controls them through its data. A process that is measured can be monitored at a distance and corrected from its records, which is why firms that base decisions on data are more productive (Brynjolfsson and McElheran, 2016) and why information technologies widen spans of control (Bloom *et al.*, 2014). Output per unit of the task bundle therefore rises with the firm's data and falls with its span of control, and I call it the firm's *control efficiency*,

$$\Psi(d, n, I) \equiv \frac{D}{S(n, I)^{\alpha}} = \frac{(d\, n)^{\eta}}{\big((1 - I)\, n\big)^{\alpha}}, \qquad 0 < \alpha < \eta < 1. \tag{11}$$

A further line adds to both the firm's data and its span of control. Since $\eta > \alpha$, the data grow faster than the span, so at given implementation control efficiency rises with the number of lines, and it rises with implementation,

$$\frac{\partial \ln \Psi}{\partial \ln n} = \eta - \alpha, \qquad \frac{\partial \ln \Psi}{\partial I} = \frac{\alpha}{1 - I}. \tag{12}$$

The records a further line generates are used on every line the firm already runs, and they improve the control of all of them by more than the wider span costs. Implementation itself rises with the number of lines, through the firm's data, which adds to the rise. Artificial intelligence takes tasks out of the span of control, so the firm's data have fewer processes to cover and every line becomes more efficient. The gain is convex in imple-

[1] Every line needs its own instance of each task type in $[0, 1]$. Appendix A.1 shows that none of the results depends on this.

mentation, since each further task removed is a larger share of what is left to control.

Firms invest in two distinct activities. Let z^q_{ijt} denote the Poisson flow rate at which research conducted from line j of firm i delivers a quality innovation, and z^d_{it} the flow rate at which the firm raises its data intensity. Each quality innovation lands on one line, while an investment in data reaches every line of the firm at once. Research combines task types in the same way as production. The firm's artificial intelligence performs the research task types it has implemented, and researchers, who work with the leading edge as every worker does, perform the rest (Appendix A.1), so a unit of the research bundle costs $B_t(I_{it})$, the unit cost of the task bundle in (27) below. The firm runs the research of all its lines from one research organisation, and innovating on a larger body of knowledge takes proportionally more research, so the firm's total rate of quality innovation, $x^q_{it} \equiv \sum_{j\in\mathcal{J}_{it}} z^q_{ijt}$, requires $\phi^q (x^q_{it})^{\vartheta^q} \bar{Q}_t$ units of the bundle. The firm's data control production and reach research only through implementation. Investment in data is also a firm-level activity, whose cost I take to be proportional to output. Attaining the two rates therefore costs the firm

$$\phi^q \big(x^q_{it}\big)^{\vartheta^q} \bar{Q}_t \, B_t(I_{it}) \qquad \text{and} \qquad \phi^d \big(z^d_{it}\big)^{\vartheta^d} Y_t, \tag{13}$$

where $\phi^q, \phi^d > 0$ govern the scale of each cost function and $\vartheta^q, \vartheta^d > 1$ their curvature, so that both are convex. The cost of research is convex in the firm's total rate of innovation rather than in the rate on each line. By (27), $\bar{Q}_t B_t(I)$ grows with the wage, and the cost of investment in data grows with output, so research effort in both activities is stationary along a balanced growth path.

Each quality innovation comes with a stochastic step λ', drawn independently across innovations and of the states of the innovator and the incumbent from the Pareto distribution

$$F(\lambda) = 1 - \lambda^{-\xi}, \qquad \lambda \geq 1, \quad \xi > 1, \tag{14}$$

and an unconditional step has

$$\mathbb{E}\big[\ln \lambda'\big] = \frac{1}{\xi}. \tag{15}$$

The step measures how far an innovation advances the codified knowledge behind the

product of the line it lands on. It decides who holds that line and, if the innovation wins the line, it is what the innovation adds to the leading edge.

A successful investment in data raises the firm's data intensity by a factor $\theta > 1$, so that

$$d_{it} = \theta^{h_{it}}, \tag{16}$$

where h_{it} is the firm's data rung. A firm enters on the first rung, $h = 1$, so that $d = \theta$ at entry, and each successful investment moves it up one rung. Records also go out of date. Processes change, and a record of how a process used to run ceases to describe how it runs now, at rate $o > 0$. A firm's data intensity therefore falls one rung at rate

$$\delta_d \equiv \frac{o}{\ln \theta}, \tag{17}$$

so that its log falls at rate o in expectation unless the firm keeps investing, and a firm on the first rung stays there. Without this, every firm that survives long enough would end on the top rung, and the data of the economy would be held by firms that no challenger can displace.

Creative destruction has two sources: incumbents that expand into new lines, and new entrants. When firm i innovates, the innovation lands on a randomly selected line j', held by firm i', and advances the codified knowledge behind that line's product by the step λ'. The two firms then compete in prices for the line. Each can supply it at its unit cost, the unit cost of the task bundle divided by control efficiency as in (31) below, and the challenger's product is better by the factor λ'. The firm with the lower quality-adjusted unit cost can price the other out of the line while still covering its own cost, so firm i displaces i' if

$$\frac{B_t(I_{it})}{\lambda' \, \Psi(d_{it}, n_{it}, I_{it})} < \frac{B_t(I_{i't})}{\Psi(d_{i't}, n_{i't}, I_{i't})}.$$

The firm that loses makes no sales on the line and leaves it, and the winner becomes its only producer. By (27) below, $B_t(I) = \left(w_t/(\gamma_\ell \bar{Q}_t)\right) b(I, \kappa_t)$, so the cost of a labour task at the leading edge is common to both sides. Data intensity follows from h and implementation

from (h, n), so the condition reads $\lambda' \, \Phi_t(h, n) > \Phi_t(h', n')$, where

$$\Phi_t(h, n) \equiv \frac{\Psi\big(d(h), n, I_t(h, n)\big)}{b\big(I_t(h, n), \kappa_t\big)} = \frac{d(h)^{\eta} \, n^{\eta - \alpha}}{\big(1 - I_t(h, n)\big)^{\alpha} \, b\big(I_t(h, n), \kappa_t\big)} \tag{18}$$

is the *composite efficiency* of a firm in state (h, n), the output it obtains on a line per unit of spending on the task bundle, measured in labour tasks at the leading edge. Composite efficiency rises with data intensity and with implementation and, since $\eta > \alpha$, with the number of lines. The contest therefore rests on price competition alone. A firm with more data and more lines supplies every line more cheaply, so its innovations win more of the lines they land on and its own lines are harder to take.

A challenger in state (h, n) therefore displaces an incumbent in state (h', n') with probability

$$\pi_t(h, n; h', n') \equiv \Pr\left(\lambda' \, \Phi_t(h, n) > \Phi_t(h', n')\right) = \min\left\{1, \left(\frac{\Phi_t(h, n)}{\Phi_t(h', n')}\right)^{\xi}\right\}. \tag{19}$$

Every step wins against an incumbent that is no more efficient than the challenger. Against a more efficient one the challenger needs a step above $\Phi_t(h', n')/\Phi_t(h, n)$, which by (14) has probability $\big(\Phi_t(h, n)/\Phi_t(h', n')\big)^{\xi}$. A larger step beats more incumbents, so the share of lines an innovation can take rises with its step. An innovation that fails to displace the incumbent is lost. One that displaces it makes the innovator the monopolist of the line, and the incumbent loses the line. Every producer uses the leading edge, so the step decides who holds the line but gives the winner no lasting advantage there.

Lemma 2 (Selected steps)**.** *Let a challenger in state (h, n) displace an incumbent in state (h', n'). Its step λ' is then Pareto with shape ξ and minimum $\pi_t(h, n; h', n')^{-1/\xi}$, the smallest winning step, so that*

$$\mathbb{E}\big[\ln \lambda' \mid \lambda' \, \Phi_t(h, n) > \Phi_t(h', n')\big] = \frac{1 - \ln \pi_t(h, n; h', n')}{\xi}. \tag{20}$$

A Pareto distribution truncated from below is Pareto with the same shape, and the

truncation point is the smallest winning step. The less likely a capture, the larger the step it carries, so lines won from incumbents much more efficient than the challenger add the most to the leading edge.

Let $\mu_t(h,n) \in [0,1]$ denote the share of lines held by firms in state (h,n). The state space is finite and discrete, with $h \in \{1,\dots,\bar{h}\}$ and $n \in \{1,\dots,\bar{n}\}$, and

$$\sum_{h=1}^{\bar{h}} \sum_{n=1}^{\bar{n}} \mu_t(h,n) = 1. \tag{21}$$

An innovation by a firm in state (h,n) therefore wins the line it lands on with probability

$$\mathbb{P}_t(h,n) = \sum_{h'=1}^{\bar{h}} \sum_{n'=1}^{\bar{n}} \mu_t(h',n')\, \pi_t(h,n;h',n'). \tag{22}$$

Since $\pi_t(h,n;h',n')$ is the probability that a step beats an incumbent in state (h',n'), $\mathbb{P}_t(h,n)$ is also the expected share of lines the firm can take. It rises with the firm's composite efficiency, and so with its data intensity and its number of lines.

New entry. There is a mass M^e of potential entrants, each of which chooses an innovation rate z^e. Entry research uses the same research tasks. A potential entrant has no data, so it implements only the public part, and the rate z^e costs $\phi^e (z^e)^{\vartheta^e} \bar{Q}_t B_t(\zeta \bar{I}_t)$, with $\phi^e > 0$ and $\vartheta^e > 1$. An entrant's innovation lands on a randomly selected line, as an incumbent's does, and the entrant starts on the first data rung with one line. An entrant therefore competes with the composite efficiency of a firm in state $(1,1)$. It displaces an incumbent in state (h',n') with probability $\pi_t(1,1;h',n')$, and its innovation succeeds with probability $\mathbb{P}_t(1,1)$, the expected share of lines an entrant can take. The first-order condition gives

$$z_t^e = \left(\frac{\mathbb{P}_t(1,1)\, V_t(1,1)}{\phi^e\, \vartheta^e\, \bar{Q}_t B_t(\zeta \bar{I}_t)} \right)^{\frac{1}{\vartheta^e - 1}}, \tag{23}$$

where $V_t(1,1)$ is the value of a one-line firm on the first rung (Section 2.2). The numerator is the expected value of an entrant's innovation: it wins a line with probability $\mathbb{P}_t(1,1)$, and the line is then worth $V_t(1,1)$ whatever step won it. By (18), a firm in state $(1,1)$ is the

least efficient in the economy, so $\mathbb{P}_t(1,1) \leq \mathbb{P}_t(h,n)$ in every state, and $\mathbb{P}_t(1,1) < 1$ as soon as any line is held by a firm in a state other than $(1,1)$. This is the barrier to entry, and the lines of the largest and most data-intensive incumbents are the hardest to enter.

Destruction rates. The contest probabilities π_t can be read from either side. Averaged over the incumbents a challenger may meet, they give its expected share of lines $\mathbb{P}_t(h,n)$ in (22). Summed over the challengers an incumbent may meet, they give the rate at which it loses its lines. An incumbent in state (h',n') loses each line at rate

$$\tau_t(h',n') = \sum_{h,n} \mu_t(h,n)\, z_t^q(h,n)\, \pi_t(h,n;h',n') + M^e z_t^e\, \pi_t(1,1;h',n'). \tag{24}$$

This is the rate at which innovations arrive on a line, $\sum_{h,n} \mu_t(h,n)\, z_t^q(h,n) + M^e z_t^e$, times the share of them that can beat an incumbent in state (h',n'), so an efficient incumbent loses its lines slowly. Averaging over incumbents gives the aggregate rate of creative destruction in terms of $\mathbb{P}_t$,

$$\tau_t \equiv \sum_{h',n'} \mu_t(h',n')\, \tau_t(h',n') = \sum_{h,n} \mu_t(h,n)\, z_t^q(h,n)\, \mathbb{P}_t(h,n) + \tau_t^e, \qquad \tau_t^e \equiv M^e z_t^e\, \mathbb{P}_t(1,1), \tag{25}$$

where τ_t^e is the rate at which entrants take lines. Each entrant brings one line, so τ_t^e is also the flow of new firms. The aggregate rate depends on the states of incumbents only through $\mathbb{P}_t$, while each incumbent's own loss rate depends on its composite efficiency. The barrier an incumbent poses to any challenger, producing firm or entrant, is its composite efficiency, which by (18) is increasing in data intensity and in the number of lines.

The leading edge. Every line that changes hands adds the log of its winning step to the log of the leading edge. An innovation by a firm in state (h,n) lands on a line held in state (h',n') with probability $\mu_t(h',n')$, wins it with probability $\pi_t(h,n;h',n')$, and then carries the selected step of Lemma 2. Summing over innovators, the leading edge evolves accord-

ing to

$$\frac{\dot{\bar{Q}}_t}{\bar{Q}_t} = \sum_{h,n} \mu_t(h,n)\, z_t^q(h,n) \sum_{h',n'} \mu_t(h',n')\, \pi_t(h,n;h',n')\, \mathbb{E}\big[\ln \lambda' \mid \lambda'\, \Phi_t(h,n) > \Phi_t(h',n')\big] + M^e z_t^e \sum_{h',n'} \mu_t(h',n')\, \pi_t(1,1;h',n')\, \mathbb{E}\big[\ln \lambda' \mid \lambda'\, \Phi_t(1,1) > \Phi_t(h',n')\big], \tag{26}$$

from the given $\bar{Q}_0$. The first term is the flow of lines that producing firms capture, each weighted by the expected log of its winning step, and the second is the same for entrants. The leading edge aggregates the log steps of the lines that change hands as aggregate log quality aggregates the improvements on individual lines in a quality-ladder model, except that the knowledge does not stay with the line.

2.2. Equilibrium

I first establish what implementation is worth to a firm, which requires task prices and is therefore an equilibrium object rather than a primitive. I then solve the static equilibrium, determining prices, sales and allocations for a given distribution of lines over firm states, and define the Markov Perfect Equilibrium of the dynamic game.

The two margins of implementation

Implementation buys two things: (i) a lower unit cost of the task bundle, in production and in research, and (ii) higher control efficiency. Constant-elasticity aggregation across tasks gives the unit cost index of a producer implementing I,

$$B_t(I) = \left[I \left(\frac{\psi}{\gamma_k} \right)^{1-\sigma} + (1-I) \left(\frac{w_t}{\gamma_\ell \bar{Q}_t} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}} = \frac{w_t}{\gamma_\ell \bar{Q}_t}\, b(I, \kappa_t), \qquad b(I,\kappa) \equiv \left[I\kappa^{1-\sigma} + 1 - I \right]^{\frac{1}{1-\sigma}}, \tag{27}$$

so that, relative to the price of a labour task, the unit cost depends only on implementation and the relative price of a machine task. The factor-cost gain from implementing one

more task type is

$$\nu(I) \equiv -\frac{\partial \ln B_t}{\partial I} = \frac{1-\kappa_t^{1-\sigma}}{(1-\sigma)\big(I\kappa_t^{1-\sigma}+1-I\big)} > 0, \tag{28}$$

which is positive under Assumption 1.

Proposition 1 (The two margins). *Let $\Phi(h,n) \equiv \Psi(d,n,I)/b(I,\kappa_t)$, with $d=\theta^h$, $D=(d\,n)^\eta$ and $I = \mathcal{A}(D)\,\bar{I}$, denote composite efficiency (18), with the time subscript dropped. At given task prices, that is at a given κ_t,*

$$\frac{\partial \ln \Phi}{\partial I} = \underbrace{\frac{\alpha}{1-I}}_{\text{control gain}} + \underbrace{\nu(I)}_{\text{factor-cost gain}}, \qquad \frac{\partial \ln \Phi}{\partial \bar{I}} = \mathcal{A}(D)\left[\frac{\alpha}{1-I}+\nu(I)\right], \tag{29}$$

and

(i) both derivatives are strictly positive, and $\partial \ln \Phi/\partial \bar{I}$ is increasing in the firm's data D;

(ii) $\dfrac{\partial^2 \ln \Phi}{\partial I^2} = \dfrac{\alpha}{(1-I)^2} + \nu'(I) > 0$, *with $\nu'(I) > 0$ because $\sigma < 1$, and*

$$\frac{\partial^2 \ln \Phi}{\partial \bar{I}^2} = \mathcal{A}(D)^2\,\frac{\partial^2 \ln \Phi}{\partial I^2} > 0;$$

(iii) $\partial^2 \ln \Phi/\partial \bar{I}\,\partial d > 0$ and $\partial^2 \ln \Phi/\partial \bar{I}\,\partial n > 0$, so the gain from an advance in the potential is increasing in data intensity and in the number of lines;

(iv) contests depend on composite efficiency only through its ratios, so a factor common to every firm's Φ cancels from every π, and wherever $\pi(h,n;h',n') < 1$,

$$\frac{\partial \ln \pi(h,n;h',n')}{\partial \bar{I}} = \xi\left[\frac{\partial \ln \Phi(h,n)}{\partial \bar{I}} - \frac{\partial \ln \Phi(h',n')}{\partial \bar{I}}\right];$$

(v) quality research costs $\phi^q(n\,z^q)^{\vartheta^q}\bar{Q}_t B_t(I)$ for the firm and entry research $\phi^e(z^e)^{\vartheta^e}\bar{Q}_t B_t(\zeta\bar{I})$, so an advance lowers a firm's cost of research at the rate $-\partial \ln B_t(I)/\partial \bar{I} = \mathcal{A}(D)\,\nu(I)$, which is increasing in the firm's data, and its cost of research relative to an entrant's at the rate $\mathcal{A}(D)\,\nu(I) - \zeta\,\nu(\zeta\bar{I}) > 0$.

Static equilibrium

The household problem yields the Euler equation

$$\frac{\dot{C}_t}{C_t} = r_t - \rho, \tag{30}$$

so that along a balanced growth path consumption and output grow at the common rate $g = r - \rho$, and the transversality condition holds because $r - g = \rho > 0$. The aggregator in (4) gives the demand schedule $y_{ijt} = Y_t\, p_{ijt}^{-\varepsilon}$, and cost minimisation over tasks gives the unit cost of the task bundle $B_t(I)$, so the marginal cost of the firm holding line j is (Appendix A.3)

$$\mathrm{MC}_{ijt} = \frac{B_t(I_{it})}{\Psi(d_{it}, n_{it}, I_{it})} = \frac{w_t/(\gamma_\ell \bar{Q}_t)}{\Phi(h,n)}. \tag{31}$$

The incumbent is the only producer of its line and faces the demand schedule above, so it sets the monopoly price, and the markup is the same on every line,

$$p_{ijt} = \frac{\varepsilon}{\varepsilon - 1}\,\mathrm{MC}_{ijt}, \qquad \chi_{ijt} = \frac{\varepsilon}{\varepsilon - 1}. \tag{32}$$

Imposing the numeraire, $\int_0^1 p_{ijt}^{1-\varepsilon}\, dj = 1$, on (32) gives the cost of a labour task at the leading edge,

$$\frac{w_t}{\gamma_\ell \bar{Q}_t} = \frac{\varepsilon - 1}{\varepsilon}\,\mathcal{E}_t^{\frac{1}{\varepsilon-1}}, \qquad \mathcal{E}_t \equiv \sum_{h=1}^{\bar{h}} \sum_{n=1}^{\bar{n}} \mu_t(h,n)\,\Phi(h,n)^{\varepsilon-1}, \tag{33}$$

where $\mathcal{E}_t$ is the economy's *efficiency index*. A line held in state (h,n) therefore has sales

$$\frac{p_{ijt} y_{ijt}}{Y_t} = s_t(h,n) \equiv \frac{\Phi(h,n)^{\varepsilon-1}}{\mathcal{E}_t}, \tag{34}$$

and, since profit is a share $1 - 1/\chi = 1/\varepsilon$ of sales, a firm in state (h,n) earns

$$\Pi_t(h,n) = \frac{n\, s_t(h,n)}{\varepsilon}\, Y_t. \tag{35}$$

Every line carries the same markup, so composite efficiency raises a firm's sales and profit, not its markup. Sales rise with composite efficiency at the elasticity $\varepsilon - 1$, rela-

tive to the efficiency index, and the sales shares add up to one, $\sum_{h,n} \mu_t(h,n)\, s_t(h,n) = 1$.

The labour cost share of the task bundle is

$$s_\ell(I) = \frac{(1-I)\left(w_t/(\gamma_\ell \bar{Q}_t)\right)^{1-\sigma}}{B_t(I)^{1-\sigma}} = \frac{1-I}{I\kappa_t^{1-\sigma} + 1 - I}, \tag{36}$$

and the rest, $1 - s_\ell(I)$, is spent on machine tasks. Composite efficiency therefore responds to the relative price of a machine task at the elasticity $\partial \ln \Phi / \partial \ln \kappa = -\big(1 - s_\ell(I)\big)$, which is largest in absolute value for the firms that implement most.

Proposition 2 (Artificial intelligence and the allocation of sales)**.** *Holding the distribution of lines fixed:*

(i) at given task prices, an advance in the potential changes the sales of a line held in state (h,n) at the rate

$$\frac{\partial \ln s_t(h,n)}{\partial \bar{I}} = (\varepsilon - 1)\left[\frac{\partial \ln \Phi(h,n)}{\partial \bar{I}} - \sum_{h',n'} \mu_t(h',n')\, s_t(h',n')\, \frac{\partial \ln \Phi(h',n')}{\partial \bar{I}}\right], \tag{37}$$

which is positive on the lines of firms whose data exceed a threshold and negative on the rest;

(ii) sales shares depend on composite efficiency only through its ratios across states, so a factor common to every firm's Φ cancels from every sales share and, by (33), raises the cost of a labour task at the leading edge in the same proportion.

By part (iii) of Proposition 1, the gain from an advance rises with the firm's data, which grow with both its data intensity and its number of lines, so an advance moves sales towards data-rich firms. Task prices respond in the same direction. The advance raises the efficiency index, and with it the cost of a labour task at the leading edge, so κ falls, and the fall raises composite efficiency most in the firms that implement most. Part (ii) is why only the dispersion of efficiency matters for the allocation of sales. What every firm gains alike is passed on to workers through the wage.

At given task prices it lowers it in two ways. Each firm's labour cost share falls as it

implements more, and by Proposition 2 sales move towards the firms that implement most, whose labour cost shares are lowest. The second is the reallocation of sales towards firms with low labour shares documented by Autor *et al.* (2020), here driven by data. The fall in κ that accompanies an advance works the other way, since with complementary tasks a cheaper machine task raises the cost share of labour.

Dynamic equilibrium

A firm is described by its data rung h and its number of lines n. A step decides whether a contest is won, but once won, a line is worth the same whatever step won it, since the step joins the leading edge. The state of a firm therefore carries no information about its lines beyond their number, and only the firm's total rate of quality innovation matters, which I write as $x^q = n\,z^q$, with z^q the rate per line. Appendix A.4 states the firm's problem at any date, (53). Along a balanced growth path Y_t, C_t and the value of every firm grow at the rate g, so with $r = \rho + g$ the value of a firm relative to output, $v(h,n) \equiv V_t(h,n)/Y_t$, satisfies

$$\begin{aligned}\rho\, v(h,n) = \max_{z^q,\, z^d} \Big\{ &\frac{n\, s(h,n)}{\varepsilon} + n\,\tau(h,n)\big[v(h,n{-}1) - v(h,n)\big] + n\, z^q\, \mathbb{P}(h,n)\big[v(h,n{+}1) - v(h,n)\big] \\ &+ z^d\big[v(h{+}1,n) - v(h,n)\big] + \delta_d\big[v(h{-}1,n) - v(h,n)\big] \\ &- \phi^q\big(n\, z^q\big)^{\vartheta^q} \frac{\omega}{\gamma_\ell}\, b\big(I(h,n),\kappa\big) - \phi^d\big(z^d\big)^{\vartheta^d} \Big\}. \end{aligned} \tag{38}$$

The terms are flow profit, the loss of each of its n lines at rate $\tau(h,n)$, expansion into a new line, investment in data, the loss of a rung as records go out of date, and investment costs, with $\bar{Q}_t B_t(I)/Y_t = (\omega/\gamma_\ell)\, b(I,\kappa)$ by (27). In the expansion term, the firm's innovations arrive at rate $n\, z^q$, and each wins the line it lands on with probability $\mathbb{P}(h,n)$. $v(h,0) = 0$, so a firm exits when it loses its last line, a firm at $\bar{h}$ or $\bar{n}$ cannot move further up in that dimension, and the rung-loss term is absent on the first rung. The expected gains from research and from investment in data are

$$\Delta^q(h,n) \equiv \mathbb{P}(h,n)\big[v(h,n{+}1) - v(h,n)\big], \qquad \Delta^d(h,n) \equiv v(h{+}1,n) - v(h,n), \tag{39}$$

and the first-order conditions give

$$n\, z^q(h,n) = \left(\frac{\max\left\{\Delta^q(h,n), 0\right\}}{\phi^q\, \vartheta^q\, (\omega/\gamma_\ell)\, b\big(I(h,n),\kappa\big)} \right)^{\frac{1}{\vartheta^q - 1}},$$
$$z^d(h,n) = \left(\frac{\Delta^d(h,n)}{\phi^d\, \vartheta^d} \right)^{\frac{1}{\vartheta^d - 1}}, \tag{40}$$

where the maximum sets quality research to zero in states in which another line would not raise the firm's value. Given the policies, (38) is one linear system on the (h,n) grid, and Appendix **??** describes how the equilibrium is computed. Entrants invest according to (23). The firm's total rate of quality innovation depends on its number of lines through Δ^q and, since the firm's data grow with its lines, through the cost of research, and the rate per line is that total divided by n. The rung affects research through both Δ^q and the cost of research. At given gains and task prices, $\partial \ln z^q / \partial \bar{I} = \mathcal{A}(D)\, \nu(I)/(\vartheta^q - 1)$ by part (v) of Proposition 1, so an advance in the potential raises quality research most in data-rich firms. Investment in data is chosen once for the firm, so its cost is borne once and its gain accrues in every line.

Assumption 3 (Limits to expansion). $(\eta - \alpha)\left(\dfrac{\varepsilon - 1}{\vartheta^q - 1} + \xi \right) < 1.$

By (18) and (35), at given implementation the flow profit of a firm in state (h,n) is proportional to $n^{1+(\eta-\alpha)(\varepsilon-1)}$, since control efficiency rises with the number of lines at the elasticity $\eta - \alpha$ and sales rise with efficiency at the elasticity $\varepsilon - 1$. A further line therefore raises flow profit by more and more, adds to the firm's data, and makes the firm's innovations more likely to win and its own lines harder to take, so expansion has increasing returns. With the cost convex in the firm's total rate of innovation, the rate per line of a large and efficient firm falls with its size. Assumption 3 is the condition under which it falls faster than the rate at which the firm loses its lines, so that a firm that grows large enough shrinks in expectation (Appendix A.4 derives it). The condition is necessary for the size of firms to be set by the economics rather than by $\bar{n}$, but not sufficient, since creative destruction itself slows as lines concentrate in efficient firms. The quantitative analysis therefore checks that few lines are held near $\bar{n}$.

Assumption 4 (Balanced growth)**.** The potential $\bar{I}$ is constant.

Lemma 3 (The balanced growth path)**.** *Under Assumption 4, let the shares of lines $\mu(h,n)$, the policies and the normalised wage ω be constant. Then the efficiency index $\mathcal{E}$ and the cost of a labour task at the leading edge $w_t/(\gamma_\ell \bar{Q}_t)$ are constant, the wage grows at the rate of the leading edge, and the relative price of a machine task is the constant solution of*

$$\ln \kappa = \ln \frac{\psi}{\gamma_k} + \ln \frac{\varepsilon}{\varepsilon - 1} - \frac{1}{\varepsilon - 1} \ln \mathcal{E}(\kappa), \tag{41}$$

where $\mathcal{E}(\kappa)$ is the efficiency index with composite efficiency evaluated at κ. The equation has at most one solution. Composite efficiency, sales shares, cost shares, research costs relative to output and the labour share are constant, and machine spending is a constant share of output, positive whenever $\bar{I} > 0$.

At every date, substituting (33) in (7) gives (41) with the efficiency index of that date, so the relative price of a machine task is pinned down by the distribution of lines (Appendix A.5 gives the proof). Labour-augmenting technical change is what makes the balanced growth path exact. Knowledge raises the productivity of workers and leaves that of machines unchanged, so the wage grows with the leading edge while the cost of a machine task keeps its ratio to the cost of a labour task (Appendix A.5).

Along a balanced growth path the shares of lines and the policies are constant, so by (26) the leading edge grows at a constant rate g. By Lemma 2 the expected log of a winning step is $(1 - \ln \pi)/\xi$, and by (22) and (25) the flow of captures is τ, so

$$g = \frac{\tau}{\xi} + \Gamma, \tag{42}$$

which by Lemma 3 is also the growth rate of the wage and of output, and where

$$\Gamma \equiv -\frac{1}{\xi} \sum_{h',n'} \mu(h',n') \Bigg[\sum_{h,n} \mu(h,n)\, z^q(h,n)\, \pi(h,n;h',n') \ln \pi(h,n;h',n') + M^e z^e\, \pi(1,1;h',n') \ln \pi(1,1;h',n') \Bigg] \geq 0 \tag{43}$$

is the selection premium. Each term is proportional to $-\pi \ln \pi$, which is zero when a contest is won for sure or never won, and positive in between. The premium is the knowledge brought in by steps large enough to displace incumbents more efficient than the innovator. Appendix A.5 derives (42) and splits it into the growth that the lines incumbents capture contribute, g^{inc}, and the growth that the lines entrants capture contribute, g^{ent}.

Definition 1 (Markov Perfect Equilibrium)**.** A Markov Perfect Equilibrium consists of an allocation $\{C_t, Y_t, y_{ijt}, k_{ijt}(x), \ell_{ijt}(x), k^R_{ijt}(x), \ell^R_{ijt}(x)\}$, with the research inputs of entrants, prices $\{r_t, w_t, p_{ijt}\}$, innovation policies $\{z^q, z^d, z^e\}$, the leading edge $\bar{Q}_t$ and the share of lines $\mu_t(h, n)$, such that: (i) the final-good producer maximises profit given prices, and the household satisfies (30) and its transversality condition; (ii) each firm implements $I_{it} = \mathcal{A}(D_{it})\bar{I}_t$, with D_{it} given by (8), and minimises cost over tasks in production and in research, giving (31) and the research costs in (13), and every entrant minimises the cost of its research with the public part implemented; (iii) the incumbent of every line sets the monopoly price (32), and lines change hands with the probabilities (19); (iv) incumbents choose z^q and z^d as functions of (h, n) to solve (53), with the expected gains (39) and policies (40), and entrants choose z^e according to (23); (v) the numeraire condition (33) holds, the labour market clears at (50), and the resource constraint (52) holds; (vi) μ_t satisfies (21) and evolves according to (54), and the leading edge evolves according to (26) from the given $\bar{Q}_0$; and (vii) along a balanced growth path, consumption, output and the wage grow at the common rate g of (42).

3. Quantitative Analysis

3.1. Calibration

Panel A of Table 1 lists the parameters I set outside the model. The discount rate is $\rho = 0.02$, and $\varepsilon = 5$ gives a markup of 1.25. Without overhead costs this already implies a profit share of 20 per cent, which is why I stay below the sales-weighted markup of 1.61 in 2016 (De Loecker *et al.*, 2020). By (27), $\sigma = 0.5$ is also the firm's elasticity of substitution between

capital and labour at given implementation, below the 0.7 estimated for US manufacturing as a whole, which includes reallocation across plants (Oberfield and Raval, 2021). I choose ψ/γ_k so that a machine task costs 27 per cent less than a labour task on the calibrated path ($\kappa = 0.73$), the average saving that Acemoglu (2025) takes from experimental studies of artificial intelligence. Research costs are cubic for quality and quadratic for data and entry, so by (40) a firm's total rate of quality innovation has an elasticity of one half with respect to its expected gain.

About 15 per cent of US worker tasks could be completed significantly faster at the same quality with a large language model alone, and 47 to 56 per cent with software built on such models (Eloundou *et al.*, 2023). Such software has to be fitted to the processes and data of the firm that uses it, so I read the first figure as the public part of the potential and the second as the whole potential, which gives $\zeta = 0.15/0.5 = 0.3$. With $\varsigma = 2$ implementation saturates with little data, and a firm with one line on the first rung already implements 91 per cent of the potential. This is consistent with Amazon's retail demand forecasts, which improve with a product's sales history at a diminishing rate and not at all with the number of products in a category (Bajari *et al.*, 2019). What an entrant lacks is therefore not a large stock of data but any data of its own.

Records go out of date at rate $o = 0.2$, a half-life of 3.5 years. Text used to train language models loses half its value in about seven years (Valavi *et al.*, 2022), and records of transactions, prices and customer behaviour are likely to date faster. Search results, for example, did not become less accurate when search engines kept their logs for shorter periods (Chiou and Tucker, 2017). The share of another firm's records that is useful, $\chi_P = 0.5$, matters only for disclosure, and Appendix B.3 reports other values of it and of o.

The span elasticity matters mainly through the managerial gain from implementation, $\partial \ln \Psi / \partial I = \alpha/(1-I)$, which no moment of the calibrated path pins down, so I set $\alpha = 0.3$. I choose $\eta - \alpha$ and the six other parameters of Panel B jointly to match seven moments, minimising the sum of squared percentage deviations with κ held at 0.73 (Appendix **??**). The targets are growth of 1.5 per cent, close to the 1.4 per cent a year at which output per hour grew in the US nonfarm business sector over 2007–2019 (U.S. Bureau of Labor

Table 1. Parameters

	Description	Value	Source or main moment
A. Set outside the model			
ρ	Discount rate	0.02	Standard
ε	Elasticity of substitution across lines	5.0	Markup of 1.25
σ	Elasticity of substitution across tasks	0.5	Below the 0.7 of Oberfield and Raval (2021)
κ	Price of a machine task relative to a labour task, calibrated path	0.73	Cost saving of 27% (Acemoglu, 2025)
ζ	Public share of the potential	0.3	15% of tasks with a model alone, 47–56% with software (Eloundou *et al.*, 2023)
α	Span elasticity	0.30	Set, see text
ς	Curvature of implementation capacity	2.0	Returns to data saturate (Bajari *et al.*, 2019)
o	Rate at which records go out of date	0.20	Half-life of 3.5 years, see text
χ_P	Share of another firm's records that is useful	0.5	Set, 0.25 and 1 in Appendix B
ϑ^q	Curvature of the cost of quality research	3	Elasticity of innovation of 1/2
ϑ^d, ϑ^e	Curvature of the costs of data and entry	2	Quadratic
M^e	Mass of potential entrants	1	Normalisation
$\bar{h},\ \bar{n}$	Top rung, largest number of lines	10, 40	Grid
B. Calibrated			
ξ	Pareto shape of quality steps	1.18	Growth rate
$\eta - \alpha$	Excess of the returns to data over the span elasticity	0.0273	Share of firms with $n \geq 4$
θ	Data intensity per rung	1.145	Mean lines per firm
$\bar{I}$	Potential	0.346	Labour share
ϕ^q	Cost of quality research	6112	R&D / sales
ϕ^d	Cost of investment in data	0.981	Investment in data / sales
ϕ^e	Cost of entry	1495	Entry rate

Statistics, 2020), a labour share of 0.60, an entry rate of 5 per cent, 2.4 lines per firm with 10 per cent of firms holding four or more, and research and development and investment in data of 3 and 2 per cent of sales.

Each target has a natural counterpart (Appendix B.1). Growth pins down ξ, because by (42) growth is the rate at which lines change hands divided by ξ, plus the selection premium. Given ξ, research intensity pins down the cost of research. The labour share pins down the potential, investment in data and entry pin down their cost scales, and the size distribution pins down θ and $\eta - \alpha$. At the calibrated values firms implement 95

Table 2. Targeted and untargeted moments

Moment	Data	Model
A. Targeted		
Growth rate of output per worker	0.0150	0.0134
Entry rate of firms	0.0500	0.0069
Mean lines per firm	2.400	2.002
Share of firms with $n \geq 4$	0.100	0.101
R&D / sales	0.0300	0.0372
Investment in data / sales	0.0200	0.0206
Labour share	0.600	0.590
B. Not targeted		
Share of firms using AI	0.180	0.913
AI use, employment-weighted / unweighted	1.778	1.033
Sales share of the largest 10% of firms	–	0.364
Entrants' share of lines changing hands	–	0.225
Selection premium / growth	–	0.036
Lines on the top data rung	–	0.014

Notes. Rates are per year. Model moments are those of the calibrated balanced growth path. The share of firms using artificial intelligence and its size gradient are from Bonney *et al.* (2026).

per cent of the potential $\bar{I} = 0.35$ on average, so machines perform a third of all tasks. Assumption 3 holds with a wide margin, and almost no lines are held near the bound of the grid, $\bar{n} = 40$.

Table 2 reports the fit. The calibrated path matches the share of firms with four or more lines, investment in data and the labour share closely, and growth, mean size and research intensity approximately. It misses the entry rate, at 0.7 against 5 per cent, for a structural reason. A firm enters only by taking a line, and with a mean log step of 0.85 growth requires lines to change hands at only 1.5 per cent a year, whereas 5 per cent entry with 2.4 lines per firm would require entrants alone to take 2.1 per cent. Smaller steps reconcile the two but make the contest so sensitive to efficiency that an advance in the potential tips the economy into concentration.

Among the moments not targeted, the largest 10 per cent of firms account for 36 per cent of sales and entrants take 23 per cent of the lines that change hands. The model overstates the share of firms using artificial intelligence, at 0.91 against 0.18, and understates its size gradient, at 1.03 against 1.8, because implementation saturates and use as firms report it has an extensive margin that the model lacks (Appendix B.1).

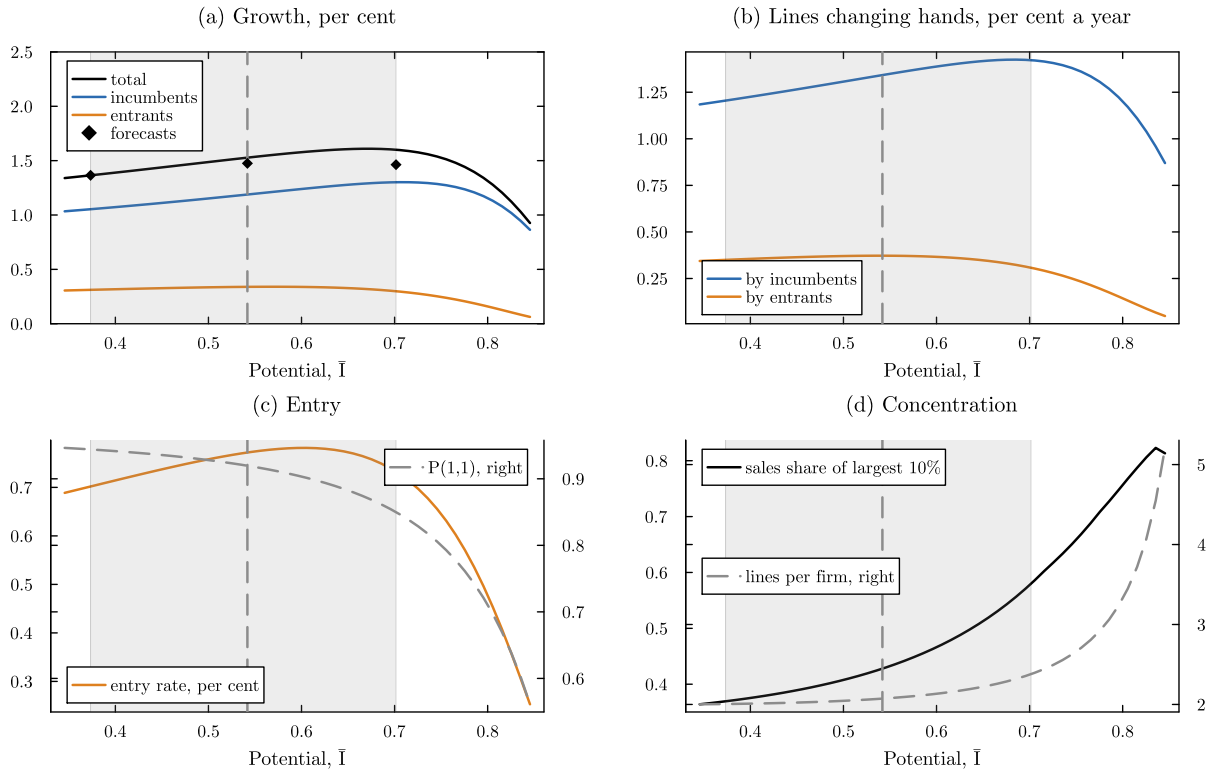


Figure 1. Balanced growth paths across the potential

Notes. Each point is a balanced growth path at the potential on the horizontal axis, with every other parameter at its calibrated value and ψ/γ_k fixed, so that κ solves (41). Growth from incumbents and from entrants is the growth contributed by the lines each group captures, (58). The shaded band runs from the potential of the pessimistic forecast to that of the optimistic forecast in Table 3, and the dashed line marks the median forecast. Diamonds in panel (a) mark growth under each forecast, including its cheaper machine tasks. $\mathbb{P}(1,1)$ is the chance that an entrant's innovation wins the line it lands on.

3.2. Artificial intelligence across balanced growth paths

Figure 1 compares balanced growth paths as the potential rises from its calibrated value of 0.35 to 0.85, with every other parameter at its calibrated value and κ moving with the wage (Table 6 in Appendix B.2 reports five of them). Beyond 0.85 more than 1 per cent of lines are held within four lines of $\bar{n}$. The shaded band is the range of current forecasts, to which I return at the end of this section.

Growth rises with the potential, but by little and not for long. It rises from 1.34 per cent to a peak of 1.61 per cent at $\bar{I} = 0.68$ and then falls, to 0.93 per cent at 0.85. Incumbents capture the whole gain. Their contribution rises from 1.03 to 1.30 percentage points by $\bar{I} = 0.70$, whereas that of entrants peaks at 0.34 points at 0.57 and falls to 0.06 points at

0.85.

Two forces of Section 2 produce this asymmetry. Research employs the task bundle, so an advance lowers the cost of research in proportion to a firm's implementation (Proposition 1, part (v)). Every incumbent implements almost the whole potential, whereas an entrant holds no records before it enters, so its research implements only the public part, $\zeta \bar{I}$. The advance also raises the composite efficiency of data-rich incumbents relative to a firm entering on the first rung with one line, so the chance that an entrant's innovation wins the line it lands on, $\mathbb{P}(1,1)$, falls from 0.95 to 0.86 at $\bar{I} = 0.70$ and to 0.56 at 0.85.

Concentration is what bounds the gain. As the advance raises the return to data, investment in data rises from 2.1 to 4.8 per cent of sales at $\bar{I} = 0.70$, and data-rich firms expand. The sales share of the largest 10 per cent of firms rises from 0.36 to 0.57, and to 0.81 at 0.85. Lines are increasingly held by firms that few challengers can displace, so the rate at which lines change hands peaks at $\bar{I} = 0.65$ and is 48 per cent lower at 0.85. The labour share falls from 0.59 to 0.40 at $\bar{I} = 0.70$, as machines take over tasks and κ falls from 0.73 to 0.29.

Where the forecasts put the economy. Rather than take a single forecast, I use the range that Aghion and Bunel (2024) derive for the next decade. Artificial intelligence takes over 4 per cent of the work of workers under their pessimistic values, which are those of Acemoglu (2025), 30 per cent under their median values and 54 per cent under their optimistic ones. Workers perform a measure $1 - \bar{I} = 0.65$ of task types on the calibrated path, so the three forecasts raise the potential by 0.03, 0.20 and 0.36.[2] The saving on the tasks taken over is 27 per cent under the pessimistic forecast, as in the calibration, and 40 per cent under the other two, for which I raise machine efficiency γ_k so that κ falls to 0.60 at the calibrated wage.

Long-run growth rises by 0.03 percentage points under the pessimistic forecast, 0.14 under the median one and only 0.12 under the optimistic one (Table 3). The larger poten-

[2]Reading the exposure shares as shares of all tasks gives advances of 0.04, 0.30 and 0.54, and the optimistic forecast then lies beyond $\bar{I} = 0.85$.

Table 3. Forecasts of artificial intelligence

	Calibrated	Pessimistic	Median	Optimistic
Share of the tasks of workers that AI takes over		0.04	0.30	0.54
Saving on the cost of a task that AI takes over	0.27	0.27	0.40	0.40
Potential, $\bar{I}$	0.35	0.37	0.54	0.70
Machine efficiency, multiple of calibrated γ_k	1	1.00	1.22	1.22
Growth rate, g (%)	1.34	1.36	1.48	1.46
change (percentage points)		+0.03	+0.14	+0.12
from more tasks		+0.03	+0.19	+0.26
from cheaper machine tasks		0.00	−0.05	−0.14
change from incumbents		+0.02	+0.14	+0.19
change from entrants		+0.01	0.00	−0.07
Entrant's chance of winning, $\mathbb{P}(1,1)$	0.95	0.94	0.90	0.82
Sales share of the largest 10% of firms	0.36	0.37	0.47	0.63
Labour share	0.59	0.57	0.50	0.43
Price of a machine task, κ	0.73	0.71	0.36	0.17

Notes. Balanced growth paths under the task-based forecasts of Aghion and Bunel (2024), whose pessimistic values are those of Acemoglu (2025). The share of the tasks of workers that artificial intelligence takes over raises the potential by that share of $1 - \bar{I}$, and a saving of 40 per cent raises machine efficiency so that κ falls to 0.60 at the calibrated wage. The change from more tasks holds machine efficiency at its calibrated value, and the change from cheaper machine tasks is the rest. Changes are relative to the calibrated path.

tial alone raises growth by 0.19 and 0.26 points under the last two, but cheaper machine tasks take back 0.05 and 0.14. Because tasks are complements, cheaper machine tasks raise the cost share of labour, which research employs, and the factor-cost gain from implementation, which favours incumbents. These are permanent changes in the growth rate, whereas the gains of 0.07 to 1.24 percentage points a year that Aghion and Bunel (2024) derive from the same forecasts are level effects spread over a decade.

3.3. Data disclosure

Incumbents gain from the advance because their records give them the implementation that entrants lack. Records are embedded knowledge. They raise the efficiency of the firm that holds them, go out of date with its processes and are lost when it exits. The knowledge in an innovation, by contrast, joins the leading edge once the innovation wins a line, and the leading edge is codified knowledge that every producer uses and that accumulates. I ask whether a rule that makes records public after a fixed number of years turns them into knowledge of the second kind.

The natural precedent is the patent, which trades exclusive rights for twenty years from filing for disclosure (Article 33 of the TRIPS Agreement), and disclosure raises follow-on innovation (Furman *et al.*, 2021; Hegde *et al.*, 2023). Records differ from inventions in three properties. They are non-rival (Jones and Tonetti, 2020), so once public they serve every firm and entrant at once. They accumulate, since public records do not leave with the firm that collected them. Finally, they date quickly. A record keeps 1.8 per cent of its value after twenty years and 37 per cent after five, so a lag as long as a patent term would make almost nothing useful public.

Under the rule, every firm deposits its records when it collects them, and they become public T years later. A public record goes out of date at rate o like any other, so along a balanced growth path the public pool holds every disclosed vintage that is still current,

$$d_P = \chi_P \, e^{-oT} \, \bar{d}, \qquad \bar{d} \equiv \sum_{h=1}^{\bar{h}} \sum_{n=1}^{\bar{n}} \mu(h,n) \, \theta^h, \tag{44}$$

where χ_P is the share of another firm's records that is useful to a firm (Appendix B.3 derives the pool from its law of motion). A firm on rung h uses $\theta^h + d_P$ in place of θ^h in (8), and an entrant implements $\mathcal{A}(d_P^{\eta})\bar{I}$ in research in place of $\zeta \bar{I}$. Without disclosure, $T = \infty$ and $d_P = 0$. The rule changes who can use records rather than who collects them, although it lowers the return to collecting them.

Once public, what a record shows about how a process runs can also be codified, and codified knowledge neither dates with the processes of the firm that collected it nor leaves with that firm. I assume that the useful content of a disclosed record joins the leading edge, as the step of an innovation does. A rung of records raises the data of the firm that collects them by the factor θ^{η}, so its content is $\eta \ln \theta$ on each of the firm's lines, and a share $\chi_P e^{-oT}$ of it is still current and useful to other firms when it becomes public. Since firms climb rungs on a share $\sum_{h,n} \mu(h,n) \, z^d(h,n)$ of lines a year, disclosed records add

$$g^P = \chi_P \, e^{-oT} \, \eta \ln \theta \sum_{h=1}^{\bar{h}} \sum_{n=1}^{\bar{n}} \mu(h,n) \, z^d(h,n) \tag{45}$$

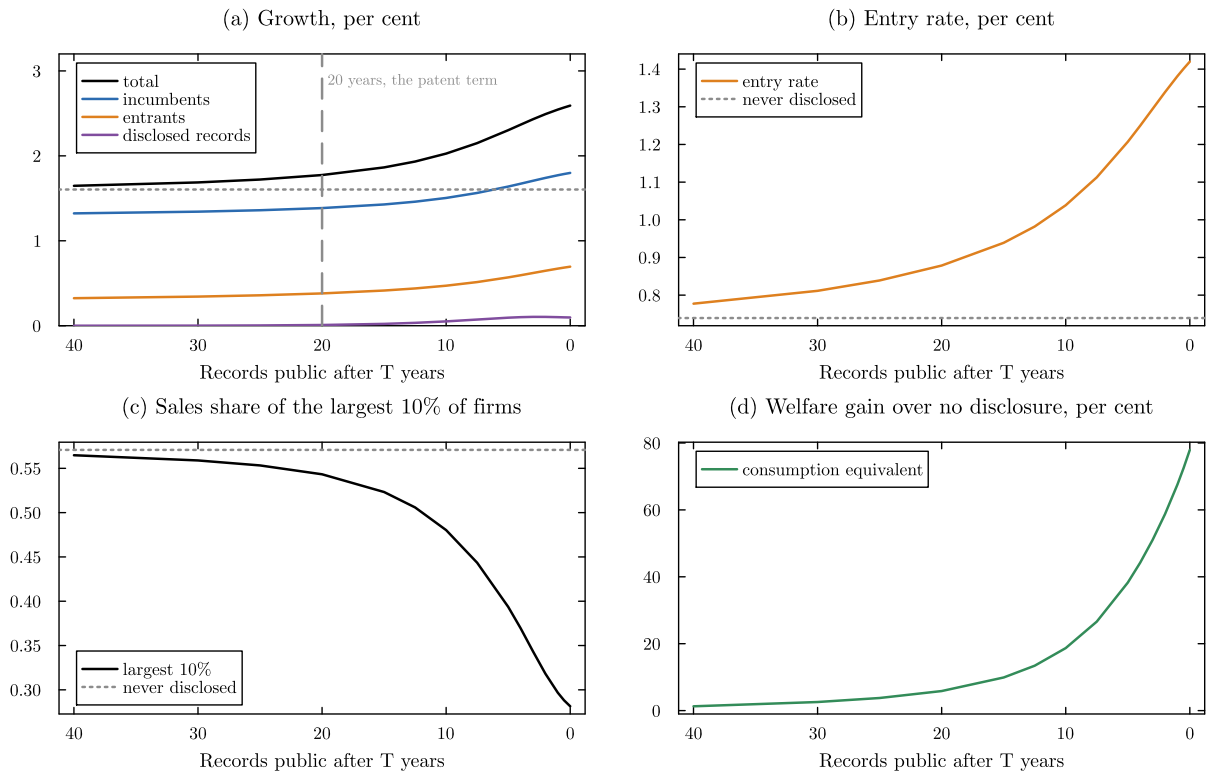


Figure 2. Data disclosure at $\bar{I} = 0.70$

Notes. Balanced growth paths with records made public after T years, at $\bar{I} = 0.70$ and every other parameter at its calibrated value. Panel (a) splits growth into the lines incumbents and entrants capture, (58), and the disclosed records that join the leading edge, (45). Dotted lines mark the path without disclosure, and the dashed line in panel (a) marks twenty years, the term of a patent. The welfare gain is (47).

to the growth of the leading edge, and the growth rate (42) becomes

$$g = \frac{\tau}{\xi} + \Gamma + g^P. \tag{46}$$

Firms take the leading edge as given, so codification changes growth but not the equilibrium, and it needs no further parameter. The pool makes records transferable and codification makes their content accumulate, which is the sense in which disclosure turns data from embedded into transferable and accumulable knowledge.

Figure 2 reports balanced growth paths at $\bar{I} = 0.70$, the potential of the optimistic forecast with machine efficiency at its calibrated value, where growth without disclosure is close to its peak, for lags from forty years to zero (Table 7 in Appendix B.3 reports six of them). Disclosure after twenty years raises growth from 1.60 to 1.77 per cent, after ten

years to 2.03 per cent and after five years to 2.30 per cent. With a ten-year lag, entry rises from 0.74 to 1.04 per cent and the sales share of the largest 10 per cent of firms falls from 0.57 to 0.48.

Most of the gain comes through research. With a ten-year lag the contribution of incumbents to growth rises from 1.30 to 1.50 percentage points and that of entrants from 0.30 to 0.47, while codified records add 0.05 points. The pool narrows the gap in efficiency between data-rich and data-poor firms, so lines that no one could take become contestable, and the distribution of firms over lines moves back towards the calibrated one (Figure 4). Shorter lags make records public while more of their value is current, but they also discourage collecting them. Investment in data falls from 4.8 per cent of sales to 3.5 per cent with a ten-year lag and to 0.25 per cent when every record is public at once, and the contribution of codified records peaks, at 0.10 points, with a lag of two to three years. Within the knowledge that disclosure codifies, this is the trade-off between appropriation and diffusion that the patent system strikes.

I measure welfare as the permanent percentage change in consumption on the path without disclosure that gives the household the utility of the path with it, with the leading edge equal at date 0. With logarithmic preferences,

$$\mathcal{W} = \exp\left(\Delta \ln c_0 + \Delta g/\rho\right) - 1, \tag{47}$$

where c_0 is consumption relative to the leading edge at date 0 (Appendix B.3). The measure ignores the transition between paths. With a ten-year lag the level of consumption falls by 4 log points and faster growth adds 21, so welfare rises by 19 per cent. The gain rises as the lag shortens, from 6 per cent at twenty years to 38 per cent at five and 78 per cent at zero, so at every lag I consider spreading records is worth more than the investment in data it discourages. The model leaves out the costs that would bound the lag from below, such as making records usable by others and protecting the privacy of the people they describe.

The case for disclosure grows with the potential, because the gap between incumbents

and an entrant's research, close to $(1-\zeta)\bar{I}$, widens as the potential rises. A ten-year lag raises welfare by 7 per cent at the calibrated potential and by 22 per cent under the full optimistic forecast, whose machine tasks are also cheaper. The gain is larger still when records last longer. With $o = 0.1$, the rate in Valavi *et al.* (2022), it is 47 per cent (Appendix B.3).

3.4. Taxes and subsidies

The usual instruments of innovation policy can also raise entry and growth. I compare disclosure with four of them: a subsidy to entry research, a subsidy to the quality research of all firms, the same subsidy restricted to small firms, with at most two lines, and a tax on investment in data. Each is settled lump sum with the household, so it changes private costs but not the resources that research and data use (Appendix B.4). Table 4 compares them with disclosure after ten years at $\bar{I} = 0.70$ at its growth rate, at its entry rate and at the setting of each instrument that maximises welfare, and Figure 3 traces each welfare gain as the setting rises.

At the growth rate of a ten-year lag, 2.03 per cent, no tax or subsidy comes close to its welfare gain of 19 per cent. An entry subsidy of 44 per cent reaches that growth rate with a welfare gain of 11 per cent, a subsidy of 53 per cent to the research of small firms with 10 per cent, a tax of 63 per cent on data with 4.5 per cent and a subsidy of 76 per cent to the research of all firms with 3 per cent. The subsidies cost between 0.9 and 11.7 per cent of output, whereas disclosure costs nothing. More generally, disclosure delivers any growth gain above about 0.17 percentage points with a larger welfare gain than any other instrument (Figure 3). Below that, a mild subsidy to the research of all firms does slightly better than lags of twenty years or more, which make almost nothing public. The disclosure frontier ends at 2.59 per cent, where every record is public at once, whereas a subsidy can always be raised further, at a falling welfare gain.

At the entry rate of a ten-year lag, 1.04 per cent, only the entry subsidy and the tax on data compete, with welfare gains of 8 and 5 per cent, since neither research subsidy reaches that rate. At their best settings the research subsidies raise welfare by 7 per cent

Table 4. Disclosure, taxes and subsidies at $\bar{I} = 0.70$

	Disclosure	Entry subsidy	R&D subsidy, all firms	R&D subsidy, small firms	Tax on data
A. Same growth rate as disclosure after ten years					
Setting	10 years	44%	76%	53%	63%
Entry rate (%)	1.04	1.14	0.68	0.85	1.01
Entrant's chance of winning	0.884	0.883	0.816	0.873	0.906
Sales share of the largest 10% of firms	0.480	0.520	0.630	0.482	0.338
Investment in data / sales (%)	3.54	4.10	5.71	3.92	0.73
Fiscal cost (% of output)	0.00	0.93	11.65	2.03	-0.46
Welfare gain (%)	18.7	11.0	2.8	10.0	4.5
B. Same entry rate as disclosure after ten years					
Setting	10 years	36%	–	–	79%
Growth rate, g (%)	2.03	1.92	–	–	2.07
Fiscal cost (% of output)	0.00	0.58	–	–	-0.28
Welfare gain (%)	18.7	8.3	–	–	4.7
C. Welfare-maximising setting					
Setting	Immediate	74%	59%	72%	113%
Growth rate, g (%)	2.59	2.64	1.87	2.32	2.11
Entry rate (%)	1.42	1.84	0.71	0.91	1.07
Fiscal cost (% of output)	0.00	6.14	5.10	5.60	-0.07
Welfare gain (%)	77.9	20.9	7.1	13.3	4.8

Notes. Small firms are firms with at most two lines. The subsidies and the tax are settled lump sum with the household. Panel A sets each instrument at the setting that gives the growth rate of disclosure after ten years, 2.03 per cent, and Panel B at the setting that gives its entry rate, 1.04 per cent. A dash marks an instrument that cannot reach that rate at any setting up to 90 per cent. Panel C reports the setting that maximises the welfare gain, which for disclosure is immediate disclosure. The fiscal cost is the subsidies paid less the taxes collected, relative to output. Welfare gains are consumption equivalents over the path without any policy, (47).

for all firms and 13 per cent for small firms, and the tax by 5 per cent. The entry subsidy does best, with 21 per cent at a rate of 74 per cent, but it needs public spending of 6.1 per cent of output, whereas disclosure after seven and a half years already raises welfare by 27 per cent at no cost, and immediate disclosure by 78 per cent.

Why does disclosure do better? At a common growth rate every instrument gains the same 21 log points of consumption from faster growth, so the ranking in Panel A is decided by the level of consumption (Table 8 in Appendix B.4), which disclosure after ten years lowers by 4 log points and the other instruments by 11 to 18. The subsidy to all firms buys growth with resources. It raises research and development to 16 per cent of sales and lowers consumption from 51 to 44 per cent of output, mostly to pay for research by the

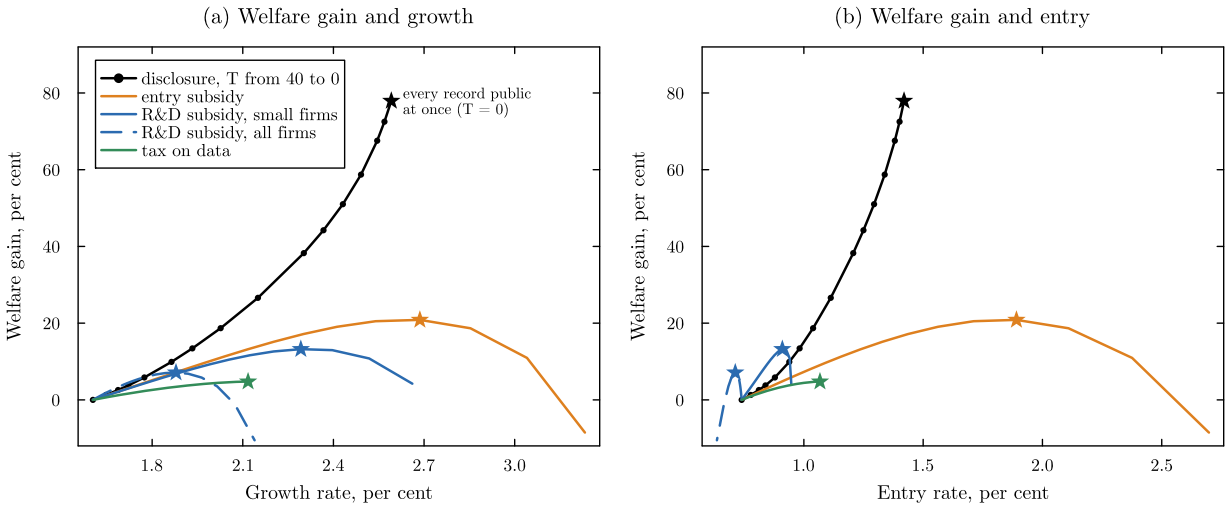


Figure 3. Welfare frontiers of disclosure, taxes and subsidies

Notes. Balanced growth paths at $\bar{I} = 0.70$. Each line traces one instrument as its setting rises from zero: the lag of disclosure falls from 40 years to zero (dots), the subsidies rise in steps of 5 percentage points to 85 or 90 per cent, and the tax on data rises to 200 per cent. Small firms are firms with at most two lines. Stars mark the setting of each instrument that maximises the welfare gain. Welfare gains are relative to the path without any policy.

firms whose records keep entrants out. The other three instruments make lines about as contestable as disclosure does but give up efficiency to get there, since the entry subsidy shortens the time a line is held, so firms invest less in data, and the tax removes records directly. The efficiency index $\mathcal{E}$ falls by 26 to 58 per cent under them, against 17 per cent under disclosure, which makes the records that firms still collect available to every firm.

With shorter lags disclosure raises the level of consumption as well, by 8 log points when every record is public at once, since the public pool raises the efficiency index above its value without disclosure. Records are non-rival, so the only cost of spreading them is the investment in data that disclosure discourages. Taxes and subsidies change how much firms invest and leave the source of the entrants' disadvantage in place, whereas disclosure changes what every firm can do with the records that exist. Financing the subsidies with distortionary taxes would widen the gap further.

4. Conclusion

Artificial intelligence can codify tacit knowledge, but what it codifies inside a firm is that firm's own knowledge. I have built an endogenous growth model in which firms implement the same technology to different degrees, governed by the data they have accumulated. Implementation substitutes capital for labour and relaxes the span of control, and both gains rise with a firm's data and its number of lines. Data stay with the firm that collects them, they go out of date, and they are lost when the firm exits, so the knowledge that artificial intelligence draws on is embedded rather than transferable.

This makes artificial intelligence a technology for incumbents. Every incumbent implements most of the potential, whereas an entrant, which holds no data, researches with the public part alone. Advances in artificial intelligence therefore raise the efficiency and the research of data-rich incumbents, make them harder to displace and concentrate sales in the largest firms, while the labour share falls. Growth rises as incumbents expand, but weaker creative destruction bounds the gain.

Requiring firms to make their records public after a few years turns data from embedded into transferable and accumulable knowledge. Public records serve every firm and every entrant as data, and what they show joins the stock of knowledge that every producer builds on. At the potential of the optimistic forecast, disclosure after ten years raises growth from 1.6 to 2.0 per cent and welfare by 19 per cent, more than any subsidy to research or entry or any tax on data achieves at the same growth rate, and at no fiscal cost.

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Appendices

A. Theoretical Appendix

A.1. Research and the span of control

The research bundle. Research assembles task types as production does. Research conducted from line j of firm i uses the bundle

$$R_{ijt} = \left(\int_0^1 r_{ijt}(x)^{\frac{\sigma-1}{\sigma}} \, dx \right)^{\frac{\sigma}{\sigma-1}}, \qquad r_{ijt}(x) = \begin{cases} \gamma_k \, k^R_{ijt}(x), & x \leq I_{it}, \\ \gamma_\ell \, \bar{Q}_t \, \ell^R_{ijt}(x), & x > I_{it}, \end{cases} \tag{48}$$

where $k^R_{ijt}(x)$ and $\ell^R_{ijt}(x)$ are the capital and labour that research task x uses. Machines perform the implemented task types, and researchers, who work with the leading edge, perform the rest. Cost minimisation is the same as in production, so a unit of the bundle costs $B_t(I_{it})$ in (27), and a total rate of quality innovation x^q_{it} requires $\phi^q (x^q_{it})^{\vartheta^q} \bar{Q}_t$ units, which gives the first cost in (13). A potential entrant holds no data, so it implements only the public part and pays $B_t(\zeta \bar{I}_t)$ per unit. The bundle carries no control-efficiency term, so data reach research only through implementation.

Tasks common to several lines. The span of control (10) assumes that every line needs its own instance of each task type, but no result depends on this. Suppose instead that a share $\beta \in (0,1)$ of every interval of task types is common to all of a firm's lines and needs controlling only once. The span of control becomes

$$S(n, I) = (1 - I)\big[\beta + (1 - \beta)\, n\big], \tag{49}$$

so implementation still enters control efficiency only through $1 - I$, $\partial \ln \Psi / \partial I = \alpha/(1-I)$ as in (12), and Proposition 1 holds as stated. At given implementation, the elasticity of

control efficiency with respect to the number of lines becomes

$$\frac{\partial \ln \Psi}{\partial \ln n} = \eta - \alpha \frac{(1-\beta)\, n}{\beta + (1-\beta)\, n} \;\geq\; \eta - \alpha,$$

which falls towards $\eta - \alpha$ as the firm grows. Control efficiency still rises with the number of lines, so an entrant is still the least efficient firm, and Assumption 3, which concerns large firms, is unchanged.

A.2. Proofs of Lemmas 1 and 2 and Proposition 1

Proof of Lemma 1. By (9), $I = \mathcal{A}(D)\,\bar{I}$ with $\mathcal{A}(D) = 1 - (1-\zeta)e^{-\varsigma D}$. (i) $\bar{I} - I = (1-\zeta)e^{-\varsigma D}\bar{I} > 0$ whenever $\bar{I} > 0$, since $\zeta < 1$ and D is finite. (ii) $\partial I/\partial \bar{I} = \mathcal{A}(D) \in (\zeta, 1)$, since $D = (d\,n)^{\eta} > 0$. Moreover $\mathcal{A}'(D) = \varsigma(1-\zeta)e^{-\varsigma D} > 0$, and by (8) $\partial D/\partial d = \eta D/d > 0$ and $\partial D/\partial n = \eta D/n > 0$, so $\mathcal{A}(D)$ increases in d and in n. (iii) Differentiating $\partial I/\partial \bar{I} = \mathcal{A}(D)$ with respect to d and to n gives $\mathcal{A}'(D)\,\eta D/d > 0$ and $\mathcal{A}'(D)\,\eta D/n > 0$. ■

Proof of Lemma 2. Let $r \equiv \Phi_t(h', n')/\Phi_t(h, n)$. The challenger wins if and only if $\lambda' > r$, and every step is at least one, so the smallest winning step is $\underline{\lambda} \equiv \max\{1, r\}$. By (14), $\Pr(\lambda' > x) = x^{-\xi}$ for $x \geq 1$, so the challenger wins with probability $\underline{\lambda}^{-\xi} = \min\{1, r^{-\xi}\}$, which is (19), and $\underline{\lambda} = \pi_t(h, n; h', n')^{-1/\xi}$. For $x \geq \underline{\lambda}$,

$$\Pr\left(\lambda' > x \mid \lambda' > \underline{\lambda}\right) = \frac{x^{-\xi}}{\underline{\lambda}^{-\xi}} = \left(\frac{\underline{\lambda}}{x}\right)^{\xi},$$

the survival function of a Pareto distribution with shape ξ and minimum $\underline{\lambda}$. Hence, conditional on winning, $\ln(\lambda'/\underline{\lambda})$ is exponential with mean $1/\xi$, and $\mathbb{E}\big[\ln \lambda' \mid \lambda' > \underline{\lambda}\big] = \ln \underline{\lambda} + 1/\xi = (1 - \ln \pi)/\xi$. With $\underline{\lambda} = 1$ this is (15). ■

Proof of Proposition 1. By (18),

$$\ln \Phi = \eta \ln d + (\eta - \alpha) \ln n - \alpha \ln(1 - I) - \ln b(I, \kappa), \qquad \ln b(I, \kappa) = \frac{\ln\left(I\kappa^{1-\sigma} + 1 - I\right)}{1 - \sigma}.$$

Holding d and n fixed, $-\partial \ln b/\partial I = \nu(I)$ in (28), which gives the first expression in (29),

and at given D, $\partial I/\partial \bar{I} = \mathcal{A}(D)$, which gives the second. Write $G(I) \equiv \alpha/(1-I) + \nu(I)$, so that $\partial \ln \Phi/\partial \bar{I} = \mathcal{A}(D)\, G(I)$. With $c \equiv 1-\kappa^{1-\sigma}$ and $x \equiv I\kappa^{1-\sigma} + 1 - I$, $\nu(I) = c/\big((1-\sigma)x\big)$ and $dx/dI = -c$, so

$$\nu'(I) = \frac{c^2}{(1-\sigma)\, x^2} = (1-\sigma)\, \nu(I)^2 > 0$$

since $\sigma < 1$, and $G'(I) = \alpha/(1-I)^2 + \nu'(I) > 0$.

(i) Under Assumption 1, $\kappa < 1$, so $c > 0$, $\nu(I) > 0$ and $G(I) > 0$, and since $\mathcal{A}(D) > 0$ both derivatives in (29) are positive. The second, $\mathcal{A}(D)\, G\big(\mathcal{A}(D)\bar{I}\big)$, is the product of two positive functions that increase in D, since $\mathcal{A}' > 0$ and $G' > 0$, so it increases in D.

(ii) $\partial^2 \ln \Phi/\partial I^2 = G'(I) > 0$, and at given D, $\partial^2 \ln \Phi/\partial \bar{I}^2 = \mathcal{A}(D)^2\, G'(I) > 0$.

(iii) The term $(\eta - \alpha)\ln n$ does not involve $\bar{I}$, so

$$\frac{\partial^2 \ln \Phi}{\partial \bar{I}\, \partial d} = \mathcal{A}'(D)\, \frac{\eta D}{d} \Big[G(I) + \mathcal{A}(D)\, \bar{I}\, G'(I) \Big] > 0,$$

and likewise for n, with $\eta D/n$ in place of $\eta D/d$.

(iv) By (19), $\pi(h, n; h', n')$ depends on $\Phi(h, n)$ and $\Phi(h', n')$ only through their ratio, so a common factor cancels. Where $\pi < 1$, $\ln \pi = \xi \big[\ln \Phi(h, n) - \ln \Phi(h', n') \big]$, and differentiating with respect to $\bar{I}$ gives the expression in the proposition.

(v) At given task prices, $B_t(I) = \big(w_t/(\gamma_\ell \bar{Q}_t)\big)\, b(I, \kappa_t)$ for production and research alike (Appendix A.1), so $-\partial \ln B_t(I)/\partial \bar{I} = \mathcal{A}(D)\, \nu(I)$, which increases in D because $\mathcal{A}$ and ν are increasing and $I = \mathcal{A}(D)\bar{I}$. An entrant implements $\zeta \bar{I}$, so its cost falls at the rate $\zeta\, \nu(\zeta \bar{I})$. Since $\mathcal{A}(D) > \zeta$ and ν is positive and increasing, $\mathcal{A}(D)\, \nu\big(\mathcal{A}(D)\bar{I}\big) > \zeta\, \nu(\zeta \bar{I})$. ■

A.3. The static equilibrium

Prices and sales. The final-good producer maximises $Y_t - \int_0^1 p_{ijt} y_{ijt}\, dj$ subject to (4), so demand is $y_{ijt} = Y_t\, p_{ijt}^{-\varepsilon}$, and the final good is the numeraire if and only if $\int_0^1 p_{ijt}^{1-\varepsilon}\, dj = 1$. An implemented task type can be performed by a machine at the unit cost ψ/γ_k or by a worker at $w_t/(\gamma_\ell \bar{Q}_t)$, and under Assumption 1 the machine is cheaper. Aggregating the task prices with elasticity σ gives $B_t(I)$ in (27), whose labour cost share is $s_\ell(I)$ in (36) by

Shephard's lemma, so that $\partial \ln \Phi / \partial \ln \kappa = -\partial \ln b / \partial \ln \kappa = -\big(1 - s_\ell(I)\big)$. Output on a line is Ψ times the bundle, which gives the marginal cost (31) and the price (32). Substituting both in the numeraire condition gives

$$1 = \Big(\frac{\varepsilon}{\varepsilon - 1}\,\frac{w_t}{\gamma_\ell \bar{Q}_t}\Big)^{1-\varepsilon} \sum_{h,\,n} \mu_t(h,n)\,\Phi(h,n)^{\varepsilon-1},$$

which is (33). Sales of a line, $p_{ijt} y_{ijt} = Y_t\, p_{ijt}^{1-\varepsilon}$, then give (34), and profit on a line is $(p_{ijt} - \mathrm{MC}_{ijt})\, y_{ijt} = p_{ijt} y_{ijt}/\varepsilon$, which summed over the firm's n lines gives (35).

Proof of Proposition 2. (i) By (34), $\ln s_t(h,n) = (\varepsilon - 1)\ln \Phi(h,n) - \ln \mathcal{E}_t$. Holding the distribution of lines fixed, (33) gives $\partial \ln \mathcal{E}_t / \partial \bar{I} = (\varepsilon - 1) \sum_{h',n'} \mu_t(h',n')\, s_t(h',n')\, \partial \ln \Phi(h',n')/\partial \bar{I}$, and subtracting gives (37). At given task prices, $\partial \ln \Phi / \partial \bar{I} = \mathcal{A}(D)\, G(\mathcal{A}(D)\bar{I})$ depends on the state only through D and, by part (i) of Proposition 1, strictly increases in D. The sales-weighted mean in (37) lies between its smallest and largest values across the states that hold lines, so there is a threshold D^* such that (37) is positive on the lines of firms with $D > D^*$ and negative on those with $D < D^*$. (ii) Multiplying every Φ by $\lambda > 0$ multiplies $\mathcal{E}_t$ by $\lambda^{\varepsilon-1}$, which leaves every sales share in (34) unchanged and, by (33), multiplies the cost of a labour task at the leading edge by λ. ■

Labour-market clearing. A line held in state (h,n) costs its sales divided by the markup, $s_t(h,n)\, Y_t(\varepsilon-1)/\varepsilon$, of which a share $s_\ell(I)$ is the wage bill. Research spends the same share on labour: $s_\ell(I_{it})$ of a firm's research cost $\phi^q (n\, z^q)^{\vartheta^q} \bar{Q}_t B_t(I_{it})$, a share $1/n$ of which falls on each of its lines, and $s_\ell(\zeta \bar{I}_t)$ of an entrant's. Labour-market clearing, $\int_0^1 \big(\ell_{ijt} + \ell^R_{ijt}\big)\, dj + M^e \ell^E_t = 1$, where ℓ_{ijt} and ℓ^R_{ijt} are the workers who produce and do research on line j and ℓ^E_t the researchers of a potential entrant, therefore gives the normalised wage

$$\omega_t \equiv \frac{w_t}{Y_t} = \sum_{h=1}^{\bar{h}} \sum_{n=1}^{\bar{n}} \mu_t(h,n)\, s_\ell\big(I(h,n)\big) \left[\frac{\varepsilon - 1}{\varepsilon}\, s_t(h,n) + \frac{\phi^q\big(n\, z^q_t(h,n)\big)^{\vartheta^q}}{n}\, \frac{\bar{Q}_t B_t\big(I(h,n)\big)}{Y_t}\right]$$
$$+\, M^e s_\ell(\zeta \bar{I}_t)\, \phi^e\big(z^e_t\big)^{\vartheta^e}\, \frac{\bar{Q}_t B_t(\zeta \bar{I}_t)}{Y_t}, \tag{50}$$

with $\bar{Q}_t B_t(I)/Y_t = (\omega_t/\gamma_\ell)\, b(I,\kappa_t)$ by (27). Since labour supply is one, ω_t is also the labour share. Research costs are proportional to ω_t, so (50) is linear in it and

$$\omega_t = \frac{W_t^{\text{prod}}}{1 - W_t^{\text{res}}}, \tag{51}$$

where W_t^{prod} is the wage bill of production relative to output and W_t^{res} that of research relative to the wage,

$$\begin{aligned} W_t^{\text{prod}} &\equiv \frac{\varepsilon - 1}{\varepsilon} \sum_{h,\,n} \mu_t(h,n)\, s_\ell\big(I(h,n)\big)\, s_t(h,n), \\ W_t^{\text{res}} &\equiv \frac{1}{\gamma_\ell} \sum_{h,\,n} \mu_t(h,n)\, s_\ell\big(I(h,n)\big)\, \frac{\phi^q\big(n\, z_t^q(h,n)\big)^{\vartheta^q}}{n}\, b\big(I(h,n),\kappa_t\big) \\ &\qquad + \frac{M^e}{\gamma_\ell}\, s_\ell(\zeta \bar{I}_t)\, \phi^e\big(z_t^e\big)^{\vartheta^e}\, b(\zeta \bar{I}_t, \kappa_t), \end{aligned}$$

which requires $W_t^{\text{res}} < 1$, so that research alone cannot employ every worker.

The resource constraint. The final good is consumed, used by machines in production and in research, or invested in data,

$$\begin{aligned} Y_t = C_t &+ \underbrace{\sum_{h,\,n} \mu_t(h,n)\Big(1 - s_\ell\big(I(h,n)\big)\Big)\, \frac{\varepsilon - 1}{\varepsilon}\, s_t(h,n)\, Y_t}_{\text{machines in production}} \\ &+ \underbrace{\sum_{h,\,n} \mu_t(h,n)\Big(1 - s_\ell\big(I(h,n)\big)\Big)\, \frac{\phi^q\big(n\, z_t^q(h,n)\big)^{\vartheta^q}}{n}\, \bar{Q}_t B_t\big(I(h,n)\big)}_{\text{machines in quality research}} \\ &+ \underbrace{M^e\big(1 - s_\ell(\zeta \bar{I}_t)\big)\, \phi^e\big(z_t^e\big)^{\vartheta^e}\, \bar{Q}_t B_t(\zeta \bar{I}_t)}_{\text{machines in entry research}} + \underbrace{\sum_{h,\,n} \mu_t(h,n)\, \frac{\phi^d\big(z_t^d(h,n)\big)^{\vartheta^d}}{n}\, Y_t}_{\text{investment in data}}. \end{aligned} \tag{52}$$

Machines take the capital share $1 - s_\ell$ of the cost of production and of research, and the wages of researchers enter (50) instead. Quality research and investment in data are spent per firm, of which there are $\mu_t(h,n)/n$ in state (h,n), and all entrants invest at the com-

mon rate z_t^e, since none knows where its innovation will land. The constraint follows from the household's budget. The household consumes its wages and the profits of firms net of research and investment in data and finances entry research, and the wages of researchers, which are both income and a research cost, cancel. By Lemma 3 the machine terms are constant shares of output along a balanced growth path, so machines, absent from Keskin (2026), keep a positive share of income.

A.4. The firm's problem and the distribution of lines

The value of a firm. At any date, the value of a firm in state (h, n) satisfies

$$\begin{aligned} r_t V_t(h,n) - \dot{V}_t(h,n) = \max_{z^q,\, z^d} \Big\{ & \Pi_t(h,n) + n\,\tau_t(h,n)\big[V_t(h,n{-}1) - V_t(h,n)\big] \\ & + n\,z^q\,\mathbb{P}_t(h,n)\big[V_t(h,n{+}1) - V_t(h,n)\big] \\ & + z^d\big[V_t(h{+}1,n) - V_t(h,n)\big] + \delta_d\big[V_t(h{-}1,n) - V_t(h,n)\big] \\ & - \phi^q\big(n\,z^q\big)^{\vartheta^q} \bar{Q}_t B_t\big(I_t(h,n)\big) - \phi^d\big(z^d\big)^{\vartheta^d} Y_t \Big\}, \end{aligned} \tag{53}$$

with $V_t(h,0) = 0$ and the terms of (38). Along a balanced growth path $V_t(h,n) = v(h,n)\,Y_t$ grows at the rate g and $r = \rho + g$ by (30), so dividing (53) by Y_t, with profit (35) and $\bar{Q}_t B_t(I)/Y_t = (\omega/\gamma_\ell)\,b(I,\kappa)$, gives (38).

First-order conditions. In (38) quality research enters through the firm's total rate $x^q = n\,z^q$, in $x^q \Delta^q(h,n) - \phi^q (x^q)^{\vartheta^q} (\omega/\gamma_\ell)\,b$, and investment in data through $z^d \Delta^d(h,n) - \phi^d (z^d)^{\vartheta^d}$. Both are strictly concave, since $\vartheta^q, \vartheta^d > 1$, so each has a unique maximiser, (40) when the gain is positive and zero otherwise. A potential entrant maximises $z^e\,\mathbb{P}_t(1,1)\,V_t(1,1) - \phi^e (z^e)^{\vartheta^e} \bar{Q}_t B_t(\zeta \bar{I}_t)$, which gives (23).

The distribution of lines. The share of lines in each state evolves according to

$$\begin{aligned}\dot{\mu}_t(h,n) = {} & z_t^d(h{-}1,n)\,\mu_t(h{-}1,n) - z_t^d(h,n)\,\mu_t(h,n) \\ & + \delta_d\,\mu_t(h{+}1,n) - \delta_d\,\mathbf{1}\{h>1\}\,\mu_t(h,n) \\ & + n\,z_t^q(h,n{-}1)\,\mathbb{P}_t(h,n{-}1)\,\mu_t(h,n{-}1) - n\,z_t^q(h,n)\,\mathbb{P}_t(h,n)\,\mu_t(h,n) \\ & + n\,\tau_t(h,n{+}1)\,\mu_t(h,n{+}1) - n\,\tau_t(h,n)\,\mu_t(h,n) + \tau_t^e\,\mathbf{1}\{(h,n)=(1,1)\}. \end{aligned} \tag{54}$$

A firm moves with all its lines. Lines enter state (h,n) when a firm one rung below raises its data intensity or one rung above loses a rung, when a firm with $n{-}1$ lines captures one, and when a firm with $n{+}1$ lines loses one. They leave when the firm holding them changes rung or captures or loses a line, with $\mu_t(\bar{h}{+}1,n)\equiv 0$. Only single-event transitions appear, since two independent Poisson events coincide with probability zero. No line leaves the grid, since $z_t^d(\bar{h},n)\equiv 0$ and $\pi_t(h,\bar{n};h',n')\equiv 0$, so that $\mathbb{P}_t(h,\bar{n})=0$. Dividing (54) by n gives the law of motion of the mass of firms in each state, $F_t(h,n)\equiv \mu_t(h,n)/n$,

$$\begin{aligned}\dot{F}_t(h,n) = {} & z_t^d(h{-}1,n)\,F_t(h{-}1,n) - z_t^d(h,n)\,F_t(h,n) + \delta_d\,F_t(h{+}1,n) - \delta_d\,\mathbf{1}\{h>1\}\,F_t(h,n) \\ & + (n-1)\,z_t^q(h,n{-}1)\,\mathbb{P}_t(h,n{-}1)\,F_t(h,n{-}1) - n\,z_t^q(h,n)\,\mathbb{P}_t(h,n)\,F_t(h,n) \\ & + (n+1)\,\tau_t(h,n{+}1)\,F_t(h,n{+}1) - n\,\tau_t(h,n)\,F_t(h,n) + \tau_t^e\,\mathbf{1}\{(h,n)=(1,1)\}. \end{aligned} \tag{55}$$

Summing over states, every move within the grid cancels, and the mass of firms rises with entry and falls with the exit of firms that lose their last line. Along a balanced growth path the two are equal,

$$\tau^e = \sum_{h=1}^{\bar{h}} \tau(h,1)\,\mu(h,1), \tag{56}$$

which the algorithm of Appendix **??** does not impose, so it checks that the fixed point has been reached.

Limits to expansion. Consider a firm on rung h that holds many lines, implements close to the potential and is more efficient than its challengers. To compare the cost

of Section 2 with that of Klette and Kortum (2004), write the cost of quality research as $\phi^q n^{1+\lambda_R(\vartheta^q-1)}(z^q)^{\vartheta^q}\bar{Q}_t B_t(I)$, which is the former for $\lambda_R = 1$ and the latter for $\lambda_R = 0$. Three elasticities with respect to the number of lines decide whether the firm keeps growing. First, at given implementation $\Phi \propto n^{\eta-\alpha}$ by (18), so flow profit, $n\, s/\varepsilon \propto n\,\Phi^{\varepsilon-1}$, rises at the elasticity $1+(\eta-\alpha)(\varepsilon-1)$, and the gain from a further line at $(\eta-\alpha)(\varepsilon-1)$. The firm wins almost every contest it enters, so $\mathbb{P}(h,n)$ is close to one and Δ^q rises at the same elasticity. Second, the first-order condition gives $z^q = n^{-\lambda_R}\big(\Delta^q/(\phi^q\vartheta^q(\omega/\gamma_\ell)b)\big)^{1/(\vartheta^q-1)}$, so the firm's rate of innovation per line changes at the elasticity $(\eta-\alpha)(\varepsilon-1)/(\vartheta^q-1)-\lambda_R$. Third, each challenger wins with probability $(\Phi_c/\Phi(h,n))^\xi$ by (19), so by (24) the rate at which the firm loses each line falls at the elasticity $\xi(\eta-\alpha)$. The expected change in the firm's lines per line it holds is $z^q\,\mathbb{P}-\tau$, so if its rate of innovation per line falls faster than its loss rate, that is if

$$(\eta-\alpha)\left(\frac{\varepsilon-1}{\vartheta^q-1}+\xi\right)<\lambda_R, \tag{57}$$

a large enough firm shrinks in expectation. With $\lambda_R = 1$, (57) is Assumption 3. With the cost of Klette and Kortum (2004) it fails whenever $\eta > \alpha$, since a larger firm then innovates faster per line and loses its lines more slowly, and the bound $\bar{n}$ of the grid ends up limiting the size of firms. The condition is necessary for the size of firms to be set by the economics rather than by $\bar{n}$, but not sufficient, since creative destruction itself slows as lines concentrate in efficient firms, and with it every firm's loss rate.

A.5. The balanced growth path

Proof of Lemma 3. Substituting (33) in (7) gives $\kappa_t = (\psi/\gamma_k)\big(\varepsilon/(\varepsilon-1)\big)\mathcal{E}_t^{-1/(\varepsilon-1)}$ at every date, and taking logs gives (41), whose right side depends on κ through composite efficiency. Since $\partial\ln\Phi/\partial\ln\kappa = -\big(1-s_\ell(I)\big)$ (Appendix A.3),

$$\frac{\partial\ln\mathcal{E}}{\partial\ln\kappa} = -(\varepsilon-1)\sum_{h,n}\mu(h,n)\,s(h,n)\,\Big(1-s_\ell\big(I(h,n)\big)\Big),$$

so the right side of (41) rises in $\ln\kappa$ with slope $\sum_{h,n}\mu(h,n)\,s(h,n)\big(1-s_\ell(I(h,n))\big)$. This slope is below one, since sales shares add up to one and $s_\ell(I)>0$ while implementation

stays below one. The difference between the two sides of (41) therefore strictly increases in $\ln \kappa$, and the equation has at most one solution. With the shares of lines constant, and $\bar{I}$ constant by Assumption 4, the right side does not depend on time, so κ is constant, and so are composite efficiency, the efficiency index and, by (33), the cost of a labour task at the leading edge. The wage therefore grows at the rate of the leading edge, and sales shares (34), cost shares (36), research costs relative to output, which are proportional to $\omega\, b(I,\kappa)$, and the labour share ω are constant. Machine tasks take a share $1 - s_\ell(I) > 0$ of every cost whenever $I > 0$, so by (52) machine spending is a constant share of output, positive whenever $\bar{I} > 0$. ■

Why technical change is labour-augmenting. Knowledge raises the productivity of workers but not of machines, so the wage grows with the leading edge while the cost of a machine task keeps its ratio to that of a labour task, which makes the balanced growth path exact. If knowledge also raised the productivity of machines, machine tasks would grow ever cheaper and, with complementary tasks, the labour cost share would tend to one, the Baumol effect in Aghion *et al.* (2019).[3] A constant machine price is consistent with balanced growth only if technical change is labour-augmenting (Jones and Scrimgeour, 2008).

The growth rate. Along a balanced growth path the shares of lines and the policies are constant. By Lemma 2, a line won with probability π carries an expected log step $(1 - \ln \pi)/\xi$, so (26) gives

$$g = g^{\text{inc}} + g^{\text{ent}}, \qquad \begin{aligned} g^{\text{inc}} &\equiv \frac{1}{\xi} \sum_{h,n} \mu(h,n)\, z^q(h,n) \sum_{h',n'} \mu(h',n')\, \pi(h,n;h',n') \big[1 - \ln \pi(h,n;h',n')\big], \\ g^{\text{ent}} &\equiv \frac{M^e z^e}{\xi} \sum_{h',n'} \mu(h',n')\, \pi(1,1;h',n') \big[1 - \ln \pi(1,1;h',n')\big], \end{aligned} \tag{58}$$

[3] Jones and Liu (2024) avoid this with an ever-expanding set of automated tasks, which is ruled out here because implementation stays below $\mathcal{A}\big((\theta^{\bar{h}}\bar{n})^\eta\big)\,\bar{I} < 1$.

the growth contributed by the lines that incumbents and entrants capture. Collecting the terms in π, by (22) and (25), gives the flow of captures τ divided by ξ, and collecting the terms in $\pi \ln \pi$ gives the selection premium (43), which is (42). By Lemma 3, g is also the growth rate of the wage and of output, since output is the wage divided by the constant ω. Under disclosure the leading edge also gains the codified content of disclosed records, the term g^P of (45).

B. Quantitative Appendix

B.1. Calibration

Identification. Table 5 reports the elasticity of each targeted moment with respect to each calibrated parameter. Every parameter moves several moments, but the matrix has a clear structure. The Pareto shape moves growth most, with an elasticity of -1.2, as (42) implies. The potential moves the labour share and little else, since it sets the share of tasks that machines perform. The cost scales of data and of entry move investment in data and the entry rate, with elasticities of -3.3 and -0.7. The parameters of the data technology, $\ln \theta$ and $\eta - \alpha$, set how much a further rung and a further line raise efficiency, and so the size distribution, and $\ln \theta$ moves every moment because it also sets how quickly firms lose rungs, $\delta_d = o/\ln \theta$. No parameter moves research intensity much, and ϕ^q moves mainly the size distribution, so research intensity is pinned down jointly with growth. Given ξ, growth fixes how often lines change hands, and ϕ^q fixes what that costs.

The use of artificial intelligence. I read reported use of artificial intelligence as the share of the firm-specific part of the potential that a firm implements, $1 - e^{-\varsigma D}$. In the United States the 18 per cent of firms that use artificial intelligence employ 32 per cent of workers (Bonney *et al.*, 2026), an employment-weighted share 1.8 times the unweighted one. With implementation saturating, the model puts the share at 0.91 and its size gradient at 1.03. Use as firms report it has an extensive margin that the model lacks, since a firm reports using artificial intelligence once it has deployed it somewhere, whereas in

Table 5. Elasticity of the targeted moments with respect to the calibrated parameters

	Growth	Entry	Lines	$n \geq 4$	R&D	Data	Labour
ξ	-1.16	-0.20	0.08	0.30	-0.24	0.97	-0.02
$\eta - \alpha$	-0.18	-0.25	0.11	0.41	-0.23	1.35	-0.02
$\ln \theta$	-1.57	-1.89	0.66	2.32	-2.48	13.85	-0.19
$\bar{I}$	0.25	0.28	-0.01	-0.02	0.16	0.24	-0.43
ϕ^q	-0.22	-0.05	-0.12	-0.44	-0.11	-0.34	-0.00
ϕ^d	0.37	0.45	-0.16	-0.59	0.59	-3.30	0.04
ϕ^e	-0.38	-0.72	0.24	0.88	-0.36	0.69	-0.03

Notes. Percentage change in each moment for a one per cent change in each parameter at the calibrated point, by central differences, with κ held at 0.73 as in the calibration. The data intensity of a rung enters through $\ln \theta$, since θ is close to one.

the model every firm implements part of the potential from the moment it enters. A calibration with shallow capacity and a fat-tailed step distribution matches both moments, at 0.18 and 1.81, but not the response of growth to the potential in Section 3.2.

B.2. Balanced growth paths across the potential

Table 6 reports five of the balanced growth paths of Figure 1. The selection premium rises from 0.05 to 0.15 percentage points at $\bar{I} = 0.85$, as the winners of contests are increasingly efficient firms, but it does not offset the fall in turnover. At high potentials incumbents themselves do less research, from 3.9 per cent of sales at 0.55 to 2.2 per cent at 0.85, because the lines they could take are held by firms they cannot displace. The mass of firms falls by three fifths, as the mean number of lines per firm rises from 2.0 to 5.2. The advance fattens the tail of the distribution of firms over lines, and disclosure after ten years moves it most of the way back (Figure 4).

B.3. Data disclosure

The public pool. A firm's data intensity summarises its stock of current records. Let c_t be the line-weighted flow of records collected at date t, so that mean data intensity is $\bar{d}_t = \int_0^\infty c_{t-a}\, e^{-\rho a}\, da$. This ignores the records that lines take with them when they change hands, which are few, since at $\bar{I} = 0.70$ lines change hands at under 3 per cent a year and records date at 20 per cent. Under the rule of Section 3.3 a record becomes public T years

Table 6. Balanced growth paths across the potential

Potential, $\bar{I}$	0.35	0.45	0.55	0.70	0.85
Growth rate, g (%)	1.34	1.43	1.53	1.60	0.93
from incumbents, g^{inc} (%)	1.03	1.11	1.19	1.30	0.86
from entrants, g^{ent} (%)	0.31	0.33	0.34	0.30	0.06
Lines taken by incumbents, τ^{inc} (%)	1.18	1.26	1.35	1.42	0.87
Lines taken by entrants, τ^{e} (%)	0.34	0.36	0.37	0.31	0.05
Selection premium, Γ (%)	0.05	0.06	0.08	0.14	0.15
Entry rate (%)	0.69	0.74	0.77	0.74	0.25
Entrant's chance of winning, $\mathbb{P}(1,1)$	0.946	0.936	0.919	0.855	0.561
Mean lines per firm	2.00	2.02	2.08	2.36	5.21
Sales share of the largest 10% of firms	0.364	0.388	0.430	0.571	0.813
Investment in data / sales (%)	2.06	2.40	2.99	4.82	7.33
R&D / sales (%)	3.72	3.85	3.93	3.74	2.24
Share of firms using AI	0.913	0.914	0.915	0.919	0.944
Labour share	0.590	0.527	0.469	0.395	0.372
Price of a machine task, κ	0.730	0.638	0.520	0.286	0.051

Notes. Each column is a balanced growth path at the potential in its head, with every other parameter at its calibrated value and ψ/γ_k fixed, so that κ solves (41). The first column is the calibrated path. Growth from incumbents and from entrants is the growth contributed by the lines each group captures, (58).

after it is collected, whatever has happened to the firm that collected it, so the pool holds the useful share χ_P of every vintage at least T years old, at the value it still has,

$$d_{P,t} = \chi_P \int_T^\infty c_{t-a}\, e^{-oa}\, da = \chi_P\, e^{-oT}\, \bar{d}_{t-T}, \qquad \dot{d}_{P,t} = \chi_P\, e^{-oT} c_{t-T} - o\, d_{P,t}. \tag{59}$$

The pool gains the records collected T years earlier, still worth e^{-oT} of their value, and loses those that date. Along a balanced growth path $\bar{d}_t$ is constant, which gives (44).

Welfare. Along a balanced growth path $C_t = c_0 \bar{Q}_0 e^{gt}$, where c_0 is consumption relative to the leading edge at date 0, so with logarithmic preferences (1) is

$$U = \int_0^\infty e^{-\rho t} \ln C_t\, dt = \frac{\ln c_0 + \ln \bar{Q}_0}{\rho} + \frac{g}{\rho^2}.$$

Raising consumption on the path without policy by the factor $1 + \mathcal{W}$ at every date raises its utility by $\ln(1+\mathcal{W})/\rho$, so with the leading edge equal at date 0 the two paths give the same utility when $\ln(1+\mathcal{W}) = \Delta \ln c_0 + \Delta g/\rho$, which is (47). Since $c_0 = (C/Y)(Y/w)(w/\bar{Q})$,

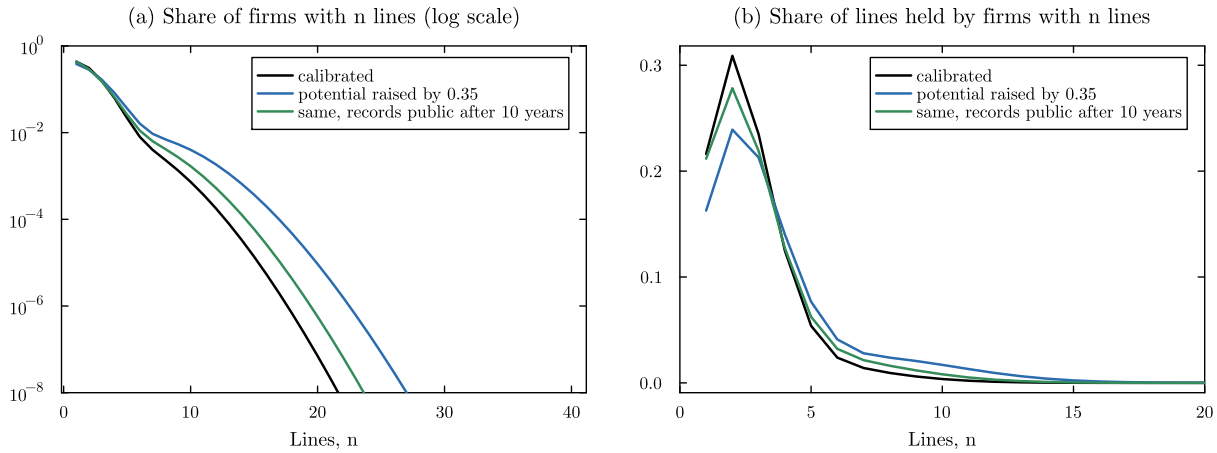


Figure 4. The distribution of firms over lines

Notes. The calibrated path, the path at $\bar{I} = 0.70$, and the same path with records made public after ten years.

(33) gives

$$\ln c_0 = \ln \frac{C}{Y} + \ln \frac{\varepsilon - 1}{\varepsilon} + \frac{\ln \mathcal{E}}{\varepsilon - 1} - \ln \omega + \ln \gamma_\ell. \tag{60}$$

The level of consumption therefore rises with the share of output that is consumed and with the efficiency index, and falls with the labour share, since at a given wage relative to the leading edge a higher labour share means that fewer workers produce. Table 8 uses this split.

Disclosure at $\bar{I} = 0.70$. Table 7 reports six of the balanced growth paths of Figure 2. Disclosure raises research and development from 3.7 to 4.6 per cent of sales after ten years and to 6.2 per cent when every record is public at once, as lines become contestable and research pays. The selection premium hardly moves, and the labour share rises slightly, since more workers do research.

Sensitivity. Figure 5 varies the rate at which records date and the usefulness of another firm's records. With $o = 0.1$, the rate in Valavi *et al.* (2022), firms accumulate more data, data-rich firms are harder to displace, and growth without disclosure at $\bar{I} = 0.70$ is only 0.73 per cent. Disclosure after ten years then raises growth to 1.73 per cent and welfare by 47 per cent, and codified records contribute 0.30 percentage points, six times as much as at $o = 0.2$, since records keep more of their value when they become public and firms

Table 7. Data disclosure at $\bar{I} = 0.70$

Records public after (years)	Never	20	10	5	1	0
Public data intensity, d_P	0.000	0.013	0.089	0.229	0.479	0.579
Growth rate, g (%)	1.60	1.77	2.03	2.30	2.54	2.59
from incumbents, g^{inc} (%)	1.30	1.39	1.50	1.64	1.77	1.80
from entrants, g^{ent} (%)	0.30	0.38	0.47	0.57	0.67	0.70
from disclosed records, g^P (%)	0.00	0.01	0.05	0.09	0.10	0.10
Lines taken by incumbents, τ^{inc} (%)	1.42	1.52	1.67	1.84	2.01	2.04
Lines taken by entrants, τ^e (%)	0.31	0.40	0.50	0.61	0.74	0.76
Selection premium, Γ (%)	0.14	0.14	0.14	0.14	0.13	0.12
Entry rate (%)	0.74	0.88	1.04	1.21	1.38	1.42
Entrant's chance of winning, $\mathbb{P}(1,1)$	0.855	0.867	0.884	0.903	0.921	0.924
Mean lines per firm	2.36	2.21	2.08	1.97	1.88	1.86
Sales share of the largest 10% of firms	0.571	0.543	0.480	0.394	0.297	0.281
Investment in data / sales (%)	4.82	4.46	3.54	2.11	0.51	0.25
R&D / sales (%)	3.74	4.06	4.58	5.27	6.03	6.17
Share of firms using AI	0.919	0.917	0.919	0.923	0.932	0.935
Labour share	0.395	0.395	0.398	0.404	0.414	0.418
Price of a machine task, κ	0.286	0.294	0.299	0.298	0.280	0.270
Welfare gain, consumption equivalent (%)	–	5.8	18.7	38.3	67.6	77.9

Notes. Balanced growth paths at $\bar{I} = 0.70$, with records made public after the number of years in the head. A lag of 0 makes every record public as soon as it is collected. Growth from disclosed records is their codified content, (45). The welfare gain is the consumption equivalent over the path without disclosure, (47).

collect more of them. With $o = 0.3$ firms invest only 0.15 per cent of sales in data, so there is little to disclose, and disclosure after ten years raises welfare by 8 per cent. With $\chi_P = 0.25$ it raises welfare by 12 per cent, and with $\chi_P = 1$ by 31 per cent. At the calibrated potential the gains are smaller: 7 per cent after ten years, 15 per cent after five and 32 per cent with immediate disclosure.

B.4. Taxes and subsidies

The instruments in the firm's problem. An entry subsidy at rate s^e replaces ϕ^e by $(1 - s^e)\phi^e$ in (23). A research subsidy at rate s^q for firms with at most n^q lines replaces ϕ^q by $(1 - s^q)\phi^q$ in their cost of quality research in (38) and in (40), with $n^q = \bar{n}$ for all firms and $n^q = 2$ for small firms. A tax at rate t^d raises the cost of investment in data to $(1 + t^d)\,\phi^d(z^d)^{\vartheta^d}$. Each instrument is settled lump sum with the household, so (50) and (52), which count the resources that research and data use, are unchanged. The fiscal cost, the

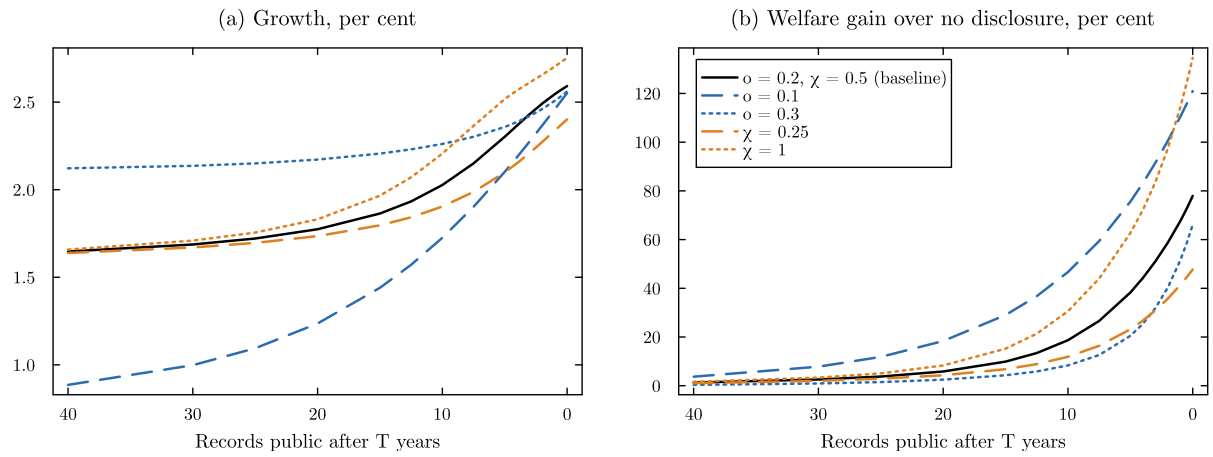


Figure 5. Sensitivity of the gain from disclosure

Notes. As Figure 2, panels (a) and (d), for other rates at which records go out of date, o, and other shares of another firm's records that are useful, χ_P, with every other parameter at its calibrated value. Each welfare gain is relative to the path without disclosure at the same o and χ_P.

subsidies paid less the tax collected, is

$$\frac{\mathcal{T}}{Y} = s^e M^e \phi^e (z^e)^{\vartheta^e} \frac{\omega}{\gamma_\ell}\, b(\zeta \bar{I}, \kappa) + s^q \sum_h \sum_{n \le n^q} \frac{\mu(h,n)}{n}\, \phi^q (n\, z^q)^{\vartheta^q} \frac{\omega}{\gamma_\ell}\, b\big(I(h,n), \kappa\big) - t^d \sum_{h,n} \frac{\mu(h,n)}{n}\, \phi^d (z^d)^{\vartheta^d}. \tag{61}$$

The sources of the welfare differences. Table 8 splits $\ln(1+\mathcal{W})$ into faster growth and the level of consumption, and the level into the three terms of (60) that move. At the growth rate of disclosure after ten years every instrument gains 21 log points from growth, so the level decides the ranking. The subsidy to the research of all firms loses through resources. Consumption falls from 51 to 44 per cent of output, 14 log points, and more workers do research, which raises the labour share and costs a further 15. The efficiency index rises by half, since most of the subsidy pays for research by data-rich incumbents, which take more lines. The other three instruments lose through efficiency. The entry subsidy raises churning, so lines are held for less time, firms invest less in data and the efficiency index falls by 28 per cent. The subsidy to small firms reaches firms only once they have entered and leaves entrants without records, and the index falls by 26 per cent. The tax removes records directly, and the index falls by 58 per cent, which the resources it saves offset only in part. Disclosure lowers the index by 17 per cent, since firms invest

Table 8. The sources of the welfare gains at $\bar{I} = 0.70$

	Disclosure, 10 years	Entry subsidy	R&D subsidy, all firms	R&D subsidy, small firms	Tax on data	Disclosure, immediate
Growth rate, g (%)	2.03	2.03	2.03	2.03	2.03	2.59
Faster growth, $\Delta g/\rho$	21.1	21.1	21.1	21.1	21.1	49.4
Level of consumption, $\Delta \ln c_0$	-4.0	-10.7	-18.4	-11.6	-16.8	8.2
consumption share, $\Delta \ln(C/Y)$	1.3	0.0	-13.8	-1.6	4.6	8.3
efficiency index, $\Delta \ln \mathcal{E}/(\varepsilon - 1)$	-4.7	-8.2	10.3	-7.5	-21.9	5.5
labour share, $-\Delta \ln \omega$	-0.6	-2.5	-14.9	-2.5	0.5	-5.6
Efficiency index, $\mathcal{E}/\mathcal{E}_0$	0.83	0.72	1.51	0.74	0.42	1.25
Welfare gain (%)	18.7	11.0	2.8	10.0	4.5	77.9

Notes. The first five columns are the settings of Panel A of Table 4, at the growth rate of disclosure after ten years, and the last is immediate disclosure. The terms are in log points, $100 \times \ln$, and add up to $\ln(1 + \mathcal{W})$ by (47) and (60): faster growth plus the level of consumption, which is the sum of the three terms below it. All are relative to the path without any policy.

less in data, but the records they still collect become available to every firm. Immediate disclosure raises the index by a quarter and the share of output that is consumed by 8 log points, since firms no longer spend on data that other firms would receive anyway.